

# HERMITIAN THEORY

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**Lemma 0.0.1.** Let  $R$  be a ring with involution (Definition 1.0.5). Suppose that there is an isomorphism  $R \cong \prod_{i=1}^n R_i$  of involution rings and write  $e_1, \dots, e_n$  for the complete central (Definition 1.0.1) orthogonal idempotents corresponding to this decomposition. Let  $\varepsilon \in Z(R)$  satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$  and write  $\varepsilon_i \in R_i$  for the component of  $\varepsilon$  in  $R_i$ . (♠ TODO: orthogonal sum)

1. Every  $\varepsilon$ -Hermitian form (Definition 1.2.1)  $H : V \times V \rightarrow R$  on a (right)  $R$ -module  $V$  uniquely decomposes as an orthogonal sum  $H = \bigoplus_{i=1}^n H_i$  where each  $H_i : V_i \times V_i \rightarrow R_i$  is an  $\varepsilon_i$ -Hermitian form on the  $R_i$ -module  $V_i = e_i V$ . In fact,  $H_i$  is constructed as  $H|_{V_i} : V_i \times V_i \rightarrow R$  and has values in  $R_i$ .
2. If  $V$  is a finitely generated projective  $R$ -module (equivalently each  $V_i$  is a finitely generated projective  $R_i$ -module), then  $H$  is nonsingular (Definition 1.2.3) if and only if each  $H_i$  is nonsingular.

*Proof.* It suffices to prove this in the case that  $R \cong R_1 \times R_2$ . We show that claimed bijection has an inverse map that is the map sending a Hermitian form  $H : V \times V \rightarrow R$  to the pair  $(H|_{V_1}, H|_{V_2})$ .

For  $i = 1, 2$ , write  $e_i$  for complete orthogonal idempotents of  $R_1 \times R_2$ . Given a  $\varepsilon$ -Hermitian form  $H : V \times V \rightarrow R$ , note that  $H|_{V_1} : V_1 \times V_1 \rightarrow R$  are respectively valued in  $R_1$  because

$$H|_{V_1}(v, w) = H(e_1 v, e_1 w) = \bar{e}_1 \cdot e_1 H(v, w) = e_1 H(v, w).$$

In fact,  $H|_{V_1}$  is an  $\varepsilon_1$ -Hermitian form over  $R_1$  because

$$H|_{V_1}(v, w) = H(e_1 v, e_1 w) = e_1 H(e_1 v, e_1 w) = e_1 \overline{\varepsilon H(e_1 w, e_1 v)} = \varepsilon_1 \overline{H|_{V_1}(w, v)}.$$

By symmetry,  $H|_{V_2} : V_2 \times V_2 \rightarrow R$  is valued in  $R_2$  and is an  $\varepsilon_2$ -Hermitian form over  $R_2$ . Moreover,  $V \cong V_1 \oplus V_2$  as  $R$ -modules, and for any  $v_1 \in V_1$  and  $v_2 \in V_2$ ,

$$H(v_1, v_2) = H(e_1 v_1, e_2 v_2) = \bar{e}_1 \cdot e_2 H(v_1, v_2) = e_1 \cdot e_2 H(v_1, v_2) = 0,$$

so  $H = H|_{V_1} \perp H|_{V_2}$ .

On the other hand, given  $H_i : V_i \times V_i \rightarrow R_i$ , we show that

$$H_i : V_i \times V_i \rightarrow R_i \hookrightarrow R_1 \times R_2 \cong R$$

is an  $\varepsilon$ -Hermitian form over  $R$ . For all  $v, w \in V_i$ ,

$$H(v, w) = \varepsilon_i \overline{H(w, v)} = \varepsilon \overline{H(w, v)}.$$

For any  $v, w \in V_i$  and  $a, b \in R$ , since  $e_i \cdot a$  and  $e_i \in b$  are identifiable as elements of  $R_i$ ,

$$H_i(av, bw) = H_i(e_i av, e_i bw) = \overline{(e_i \cdot a)} H_i(v, w) (e_i \cdot b) = \bar{a} H_i(v, w) b,$$

so  $H_i$  is indeed a Hermitian form over  $R$ . Note that  $(H_1 \perp H_2)_{V_i} = H_i$ , so the claimed inverse map is indeed the inverse map to the claimed bijection as desired. By ??, the  $H_i$  are all nonsingular if and only if  $\perp_i H_i$  is nonsingular.  $\square$

## 1. SESQUILINEAR AND HERMITIAN FORMS

**Definition 1.0.1** (Center of a ring). Let  $R$  be a (not necessarily commutative) ring. The *center of  $R$* , denoted by  $Z(R)$ , is the subset of elements in  $R$  that commute with every element of  $R$ :

$$Z(R) := \{z \in R \mid zr = rz \text{ for all } r \in R\}.$$

The center  $Z(R)$  is a commutative subring of  $R$ .

Elements of  $Z(R)$  are called *central elements of  $R$* .

### 1.0.1. *Involution ring.* (♠ TODO: don't want involution; want dagger objects)

**Definition 1.0.2.** Let  $\mathcal{C}$  be a category.

A *dagger object in  $\mathcal{C}$*  is an object  $X$  equipped with a functor  $(\cdot)^* : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$

together with an isomorphism  $i : X \rightarrow X^*$  such that the following diagram commutes:

$$X \xrightarrow{i} X^* \xrightarrow{i^*} X^{**} \cong X = \text{id}_X$$

is an automorphism  $i : X \rightarrow X$  such that  $i \circ i = \text{id}_X$ .

**Definition 1.0.3.** Let  $\mathcal{C}$  be a category. An *involution on an object  $X \in \text{Ob}(\mathcal{C})$*  is an automorphism  $i : X \rightarrow X$  such that  $i \circ i = \text{id}_X$ .

**Definition 1.0.4.** Let  $\mathcal{C}$  be a category. The *trivial involution on an object  $C$  of  $\mathcal{C}$*  simply refers to the identity map on  $C$ ; note that it is an involution.

**Definition 1.0.5.** Let  $R$  be a unital associative ring. An *involution on  $R$*  is a anti-ring homomorphism of order 2, i.e. a map  $\sigma : R \rightarrow R$  satisfying the following conditions for all  $a, b \in R$ :

1.  $\sigma(a + b) = \sigma(a) + \sigma(b)$ ;
2.  $\sigma(ab) = \sigma(b)\sigma(a)$  (anti-homomorphism property);
3.  $\sigma(1) = 1$ ;
4.  $\sigma(\sigma(a)) = a$ .

In many contexts, an involution may simply be notated by  $\cdot \mapsto \bar{\cdot}$ . A *ring with involution* is a ring equipped with an involution.

Equivalently, a ring with involution is a dagger object (Definition 1.0.2) in the category of rings.

**Definition 1.0.6.** A *homomorphism/morphism  $\varphi : (A, \sigma_A) \rightarrow (B, \sigma_B)$  of involution rings* is a ring homomorphism  $\varphi : A \rightarrow B$  such that  $\sigma_B(\varphi(a)) = \varphi(\sigma_A(a))$  for all  $a \in A$ .

In particular, if  $(A, \sigma_A)$  is an involution ring, then an *involution algebra* is an involution ring  $(B, \sigma_B)$  equipped with a homomorphism  $\varphi : (A, \sigma_A) \rightarrow (B, \sigma_B)$  of involution rings such that  $\varphi$  makes  $B$  into an  $A$ -algebra, i.e.  $\varphi$  maps  $A$  into the center (Definition 1.0.1) of  $B$ .

**Definition 1.0.7.** Let  $R$  be a ring.

The *trivial involution on  $R$*  refers to the identity ring homomorphism on  $R$ . (🔥 **TODO: have to change this to a discussion of daggers, not categorical involutions**) Equivalently, the trivial involution on  $R$  is simply the trivial involution (Definition 1.0.4) on  $R$  considered as an object of the category of rings.

**Definition 1.0.8.** 1. Let  $(R, \sigma_R)$  be a ring with involution (Definition 1.0.5). An  *$R$ -algebra with involution* is an  $R$ -algebra  $A$  equipped with an involution (Definition 1.0.5)  $\sigma_A$ , which may be called an  *$R$ -linear involution*, such that:

$$\sigma_A(ra) = \sigma_A(a)\sigma_R(r) \quad \text{for all } r \in R, a \in A.$$

(Equivalently,  $\sigma_A(ar) = \sigma_R(r)\sigma_A(a)$ ).

2. Let  $R$  be a ring.

An  *$R$ -algebra with  $R$ -linear involution* or simply an  *$R$ -algebra with involution* usually refers to an  $R$ -algebra with involution in the above sense where  $R$  is considered as an involution ring equipped with the trivial involution (Definition 1.0.7).

## 1.1. Sesquilinear forms on modules over involution rings.

**Definition 1.1.1.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5).

1. Let  $M$  be a right  $A$ -module. A  $\sigma$ -sesquilinear form (or simply sesquilinear form) on  $M$  is a map  $b : M \times M \rightarrow A$  such that

(a)  $b$  is bi-additive, i.e.

$$b(u + v, w) = b(u, w) + b(v, w) \text{ and } b(u, v + w) = b(u, v) + b(u, w)$$

for all  $u, w, \in M$ , and

(b)  $b$  is conjugate-linear in the first variable and linear in the second variable, i.e.

$$b(ua_1, va_2) = \sigma(a_1) \cdot b(u, v) \cdot a_2$$

for all  $u, v \in M$  and  $a_1, a_2 \in A$ .

2. Let  $M$  be a left  $A$ -module. A  $\sigma$ -sesquilinear form (or simply sesquilinear form) on  $M$  is a map  $b : M \times M \rightarrow A$  such that

(a)  $b$  is bi-additive, i.e.

$$b(u + v, w) = b(u, w) + b(v, w) \text{ and } b(u, v + w) = b(u, v) + b(u, w)$$

for all  $u, w, \in M$ , and

(b)  $b$  is linear in the first variable and conjugate-linear in the second variable, i.e.

$$b(a_1u, a_2v) = a_1 \cdot b(u, v) \cdot \sigma(a_2)$$

for all  $u, v \in M$  and  $a_1, a_2 \in A$ .

Note that the difference between sesquilinear forms on right  $A$ -modules vs. left  $A$ -modules is which variable is linear and which variable is conjugate-linear. In case the module is both a left and a right  $A$ -module, one should more directly emphasize which variable is linear and which is conjugate-linear. [?] discusses sesquilinear forms on right  $A$ -modules. We usually adopt this same convention by default.

We may denote the set of sesquilinear forms on  $M$  by  $\text{Sesq}_A(M)$ . It is a module over the center of  $A$  (Definition 1.0.1).

**Definition 1.1.2.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5). The construction  $M \mapsto \text{Sesq}_A(M)$  (Definition 1.1.1) is functorial; there is a contravariant functor

$$\text{Sesq}_A : \text{Mod}_A^{\text{op}} \rightarrow {}_{Z(A)}\text{Mod}$$

(Definition 1.0.1) that sends a homomorphism  $\varphi : M_1 \rightarrow M_2$  of  $A$ -modules to the  $Z(A)$ -linear map

$$\text{Sesq}_A(\varphi) : \text{Sesq}_A(M_2) \rightarrow \text{Sesq}_A(M_1), \quad \text{Sesq}_A(\varphi)(b)(x, y) = b(\varphi(x), \varphi(y)).$$

Equivalently, the corresponding map

$$\text{Hom}_A(M_2, M_2^*) \rightarrow \text{Hom}_A(M_1, M_1^*)$$

for the adjoints is given by  $h \mapsto \varphi^* h_\varphi$  (Definition 1.1.14).

**Definition 1.1.3.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5). Let  $(M_1, b_1)$  and  $(M_2, b_2)$  be (right) modules equipped with sesquilinear forms (Definition 1.1.1). A *morphism*  $(M_1, b_1) \rightarrow (M_2, b_2)$  between the sesquilinear modules is a (right)  $A$ -module homomorphism such that  $\text{Sesq}(\varphi)(b_2) = b_1$ , i.e.

$$b_2(\varphi(x), \varphi(y)) = b_1(x, y)$$

for all  $x, y \in M_1$ .

**Definition 1.1.4.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5). Let  $(M_1, b_1)$  and  $(M_2, b_2)$  be (right) modules equipped with sesquilinear forms (Definition 1.1.1).

An *isometry*  $\varphi : (M_1, b_1) \rightarrow (M_2, b_2)$  is a morphism of sesquilinear modules (Definition 1.1.3) that is an isomorphism. If an isometry exists between the two sesquilinear modules, then the two forms are said to be *isometric*.

**Definition 1.1.5.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5).

The *category of modules with sesquilinear forms over  $(A, \sigma)$*  is the category whose objects are pairs  $(M, b)$  where  $M$  is a (right)  $A$ -module and  $b : M \times M \rightarrow A$  is a sesquilinear form (Definition 1.1.1) and whose morphisms are isometries (Definition 1.1.4).

In particular, such a category is a groupoid.

**Definition 1.1.6.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be (right) module equipped with a sesquilinear form (Definition 1.1.1). The isometries (Definition 1.1.4) of  $(M, b)$  form a group with respect to composition called the *orthogonal* or *unitary group of  $(M, b)$* . The group may be denoted by notations such as  $O(M, b)$ ,  $O_A(M, b)$ ,  $U(M, b)$ , and  $U_A(M, b)$ .

#### 1.1.1. Twisted dual modules on involution rings.

**Definition 1.1.7.** Let  $R$  be a (not necessarily commutative) ring. Depending on the module structure of  $M$ , we define its dual module as follows:

1. If  $M$  is a left  $R$ -module, then the *(right) dual module of  $M$*  is

$$M^* = M^\vee := \text{Hom}_R(M, R) = \text{Hom}_{R-\mathbb{Z}}(M, R).$$

Note that it is a right  $R$ -module, as  $M$  is a  $R-\mathbb{Z}$ -bimodule and  $R$  is an  $R-R$ -bimodule.

2. If  $M$  is a right  $R$ -module, then the *(left) dual module of  $M$*  is

$${}^*M = {}^\vee M := \text{Hom}_R(M, R) = \text{Hom}_{\mathbb{Z}-R}(M, R).$$

Note that it is a left  $R$ -module, as  $M$  is a  $\mathbb{Z}-R$ -bimodule and  $R$  is an  $R-R$ -bimodule.

3. If  $M$  is a two-sided  $R$ -module, then the *dual of  $M$*  usually refers to either the right or the left dual as above.

If a convention for the rightness or the leftness of modules is fixed, then the appropriate dual module is often denoted by  $M^\vee$ .

In any case, the functor  $M \mapsto M^\vee$  is a contravariant functor from the appropriate category of modules to itself. More explicitly, the dual module constructions yield functors

$$\begin{aligned} (-)^\vee : {}_R \text{Mod}^{\text{op}} &\rightarrow \text{Mod}_R \\ (-)^\vee : \text{Mod}_R^{\text{op}} &\rightarrow {}_R \text{Mod}. \end{aligned}$$

If  $R$  is a field  $F$  and  $V$  is an  $F$ -vector space, then the dual module

$$V^* = V^\vee := \text{Hom}_F(V, F)$$

is called the *dual vector space of  $V$* .

**Definition 1.1.8.** Let  $R$  be a (not necessarily commutative) ring. Let  $M$  be a (left/right)  $R$ -module.

The *canonical (evaluation) map from  $M$  to its double dual* refers to the morphism

$$M \rightarrow (M^\vee)^\vee = \text{Hom}_R(M^\vee, R) = \text{Hom}_R(\text{Hom}_R(M, R), R), \quad m \mapsto (f \mapsto f(m)).$$

**Definition 1.1.9.** Let  $R$  be a ring with involution (Definition 1.0.5)  $r \mapsto \bar{r}$ .

Let  $M$  be a left/right  $R$ -module. The *opposite module* (not to be confused with the notion of opposite modules for modules over general rings), also called the *conjugate module*, is the right/left  $R$ -module, denoted by notations such as  $M^{op}$  or  $\bar{M}$ , with the same underlying abelian group as  $M$ , where the right  $R$ -action is defined as follows:

•

$$m \cdot r = \bar{r}m$$

for all  $m \in M$  and  $r \in R$  if  $M$  is a left  $R$ -module.

$$r \cdot m = m\bar{r}$$

for all  $m \in M$  and  $r \in R$  if  $M$  is a right  $R$ -module.

The opposite/conjugate module construction yields functors

$$\begin{aligned} \overline{(-)} : {}_R \text{Mod} &\rightarrow \text{Mod}_R \\ \overline{(-)} : \text{Mod}_R &\rightarrow {}_R \text{Mod}. \end{aligned}$$

**Proposition 1.1.10.** Let  $R$  be a ring with involution (Definition 1.0.5). The opposite module constructions provide an equivalence of categories between the category of left  $R$ -modules and the category of right  $R$ -modules. In fact, if  $M$  is a left/right  $R$ -module, then  $(M^{op})^{op}$  is naturally isomorphic to  $M$  as a left/right  $R$ -module via the identity map.

**Definition 1.1.11.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5). Let  $N$  be a one-sided  $A$ -module. The *twisted dual module*, denoted by notations such as  $N^*$ , (or sometimes  $N^\vee$ ; not to be confused with the standard dual (Definition 1.1.7)  $N^\vee$ ) or  $\text{Hom}_A(N, A)_\sigma$ , is the following:

1. In the case that  $N$  is a right  $A$ -module, it is the set of all right  $A$ -linear maps  $f : N \rightarrow A$ , equipped with the right  $A$ -module structure given by

$$(f \cdot r)(n) = \sigma(r) \cdot f(n)$$

for all  $r \in A$ ,  $f \in N^\vee$ ,  $n \in N$ . Equivalently, it may be constructed as  $\overline{N^\vee}$ , where by  $\bar{\cdot}$ , we mean the opposite module (Definition 1.1.9) and by  $\cdot^\vee$ , we mean the standard dual (Definition 1.1.7).

2. In the case that  $N$  is a left  $A$ -module, it is the set of all left  $A$ -linear maps  $f : N \rightarrow A$ , equipped with the left  $A$ -module structure given by

$$(r \cdot f)(n) = f(n) \cdot \sigma(r)$$

for all  $r \in A$ ,  $f \in N^\vee$ ,  $n \in N$ . Equivalently, it may be constructed as  $\overline{N^\vee}$ , where by  $\overline{\cdot}$ , we mean the opposite module (Definition 1.1.9) and by  $\cdot^\vee$ , we mean the standard dual (Definition 1.1.7).

Twisted dual module constructions thus yield functors

$$\begin{aligned} \text{Mod}_A^{\text{op}} &\rightarrow \text{Mod}_A \\ {}_A \text{Mod}^{\text{op}} &\rightarrow {}_A \text{Mod}. \end{aligned}$$

Given a morphism  $h : M \rightarrow N$  of right or left  $A$ -modules, we may write  $h^*$  for the morphism  $h^* : N^* \rightarrow M^*$  in functorial fashion.

The construction  $N^* = \overline{N^\vee}$  is naturally isomorphic to  $(\overline{N})^\vee$  (Lemma 1.1.12).

**Lemma 1.1.12** ([?, Chapter I, 2.1.1 Lemma]). Let  $A$  be a ring with involution (Definition 1.0.5) denoted by  $a \mapsto \bar{a}$ .

For (left/right)  $A$ -modules  $M$ , the maps

$$T_M : \overline{M^\vee} \rightarrow \overline{M}^\vee$$

(Definition 1.1.7) (Definition 1.1.9) given by

$$f \mapsto \left( T_M(f) : \overline{m} \mapsto \overline{f(m)} \right)_s$$

where  $\overline{m} = m$  as an element over  $\overline{M}$ , defines a natural isomorphism

$$\overline{(-)^\vee} \rightarrow (\overline{-})^\vee$$

between the functors  $\overline{(-)^\vee}, (\overline{-})^\vee$  from the category of (left/right)  $A$ -modules to itself.

In other words, the twisted dual module (Definition 1.1.11)  $M^*$  may be constructed either as  $\overline{M^\vee}$  or as  $\overline{M}^\vee$ .

**Definition 1.1.13** ([?, Chapter I, After Lemma 2.1.1]). Let  $A$  be a ring with involution (Definition 1.0.5) denoted by  $a \mapsto \bar{a}$ .

Given a left/right  $A$ -module  $M$ , one can speak of the double (twisted) dual module (Lemma 1.1.12)  $(M^*)^*$ .

By Lemma 1.1.12, there is a canonical isomorphism

$$\overline{T_{M^\vee}} : \overline{(M^\vee)^\vee} \rightarrow M^{**}$$

(Definition 1.1.7) (Definition 1.1.9). Define the *canonical morphism from  $M$  to its double (twisted) dual* as the composition

$$\text{can}_M : M \rightarrow (M^\vee)^\vee \cong \overline{(M^\vee)^\vee} \xrightarrow{\overline{T_{M^\vee}}} M^{**}$$

where  $M \rightarrow (M^\vee)^\vee$  is the canonical evaluation map (Definition 1.1.8).

More explicitly,  $\text{can}_M(m) = m^{**} \in M^{**}$ , where  $m^{**}(f) = \overline{f(m)}$ . Moreover,  $\text{can}_M$  specifies a natural transformation from the identity functor to the double dual functor on the appropriate category of  $A$ -modules:

$$\begin{aligned}\text{can}_M : \text{id}_{{}_A\text{Mod}} &\Rightarrow (-)^{**} : {}_A\text{Mod} \rightarrow {}_A\text{Mod} \\ \text{can}_M : \text{id}_{\text{Mod}_A} &\Rightarrow (-)^{**} : \text{Mod}_A \rightarrow \text{Mod}_A\end{aligned}$$

1.1.2. *Sesquilinear forms  $M \times M \rightarrow A$  are equivalent to maps  $M \rightarrow M^*$ .*

**Definition 1.1.14** ([?, Chapter I, 2.2]). Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5) and  $M$  a right/left  $A$ -module equipped with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow A$ .

1. The *left adjoint map* is the assignment

$$\hat{b}_l : u \mapsto (v \mapsto b(u, v)).$$

In the case that  $M$  is a right  $A$ -module, it is a homomorphism morphism of right  $A$ -modules:

$$\hat{b}_l : M \rightarrow M^*, \quad u \mapsto (v \mapsto b(u, v))$$

where  $M^* = \text{Hom}_A(M, A)_\sigma$  is the twisted dual module (Definition 1.1.11).

2. The *right adjoint map* is the assignment

$$\hat{b}_r : v \mapsto (u \mapsto b(u, v)).$$

In the case that  $M$  is a left  $A$ -module, it is a homomorphism morphism of left  $A$ -modules:

$$\hat{b}_r : M \rightarrow M^*, \quad v \mapsto (u \mapsto \sigma(b(u, v)))$$

When we simply talk about the *adjoint map* of  $b$  on the right/left  $A$ -module  $M$ , we will mean the left/right adjoint respectively and denote it by  $\hat{b}$  or by  $h_b$ .

**Lemma 1.1.15.** Let  $A$  be an involution ring (Definition 1.0.5), and let  $M$  be a left/right  $A$ -module. Let  $b : M \times M \rightarrow A$  be a sesquilinear form (Definition 1.1.1)

1. Suppose that  $M$  is a left  $A$ -module. The right adjoint map (Definition 1.1.14)  $\hat{b}_r$  is indeed a left  $A$ -module homomorphism

$$M \rightarrow M^* = \text{Hom}_{{}_A\text{Mod}}(M, A)_\sigma.$$

(Definition 1.1.11)

2. Suppose that  $M$  is a right  $A$ -module. The left adjoint map (Definition 1.1.14)  $\hat{b}_l$  is indeed a right  $A$ -module homomorphism

$$M \rightarrow M^* = \text{Hom}_{\text{Mod}_A}(M, A)_\sigma.$$

(Definition 1.1.11)

*Proof.* 1. Suppose that  $M$  is a left  $A$ -module. For all  $v \in M$ , the map

$$\widehat{b}_r(v) = b(-, v) : u \mapsto b(u, v).$$

is a left  $A$ -linear map because  $b$  is assumed to be left  $A$ -linear in the first argument. For all  $v \in M$  and  $a \in A$ , and regarding  $\widehat{b}_r(v)$  as an element of  $M^*$  with its left  $A$ -module structure, we have

$$\widehat{b}_r(av)(u) = b(u, av) = b(u, v) \cdot \bar{a} = (a \cdot \widehat{b}_r(v))(u).$$

Therefore,  $\widehat{b}_r$  is indeed a left  $A$ -module homomorphism.

2. We proceed symmetrically. Suppose that  $M$  is a right  $A$ -module. For all  $u \in M$ , the map

$$\widehat{b}_l(u) = b(u, -) : v \mapsto b(u, v).$$

is a right  $A$ -linear map because  $b$  is assumed to be right  $A$ -linear in the second argument. For all  $u \in M$  and  $a \in A$ , and regarding  $\widehat{b}_l(u)$  as an element of  $M^*$  with its right  $A$ -module structure, we have

$$\widehat{b}_l(ua)(v) = b(ua, v) = \bar{a} \cdot b(u, v) = (\widehat{b}_l(u) \cdot a)(v).$$

Therefore,  $\widehat{b}_l$  is indeed a right  $A$ -module homomorphism.

□

**Definition 1.1.16.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $M$  be a left/right  $A$ -module. Let  $h : M \rightarrow M^*$  be a left/right  $A$ -module homomorphism where  $M^*$  is the twisted dual module of  $M$  (Definition 1.1.11). The *sesquilinear form associated to  $h$*  is the sesquilinear form (Definition 1.1.1)  $b_h : M \times M \rightarrow A$  given by

1.  $b_h(u, v) = h(v)(u)$  in the case that  $M$  is a left  $A$ -module.
2.  $b_h(u, v) = h(u)(v)$  in the case that  $M$  is a right  $A$ -module.

It is indeed a sesquilinear form (Lemma 1.1.17).

**Lemma 1.1.17.** Let  $A$  be an involution ring (Definition 1.0.5), and let  $M$  be a left/right  $A$ -module. Let  $h : M \rightarrow M^*$  be a left/right  $A$ -module homomorphism where  $M^*$  is the twisted dual module of  $M$  (Definition 1.1.11). The sesquilinear form  $b_h$  associated to  $h$  (Definition 1.1.16) is indeed a sesquilinear form (Definition 1.1.1) over  $A$ .

*Proof.* Suppose that  $M$  is a left  $A$ -module. Since  $h$  is a left  $A$ -module homomorphism,  $h(av) = a \cdot h(v)$  for all  $a \in A$  and  $v \in M$ . Thus,

$$b_h(u, av) = h(av)(u) = (ah(v))(u) = h(v)(u) \cdot \bar{a}$$

for all  $u \in M$ . Moreover, for all  $a \in A$  and  $u \in M$ , since  $h(v) \in M^* = \text{Hom}_A(M, A)_\sigma$  is a left  $A$ -module homomorphism,

$$b_h(au, v) = h(v)(au) = a \cdot (h(v)(u)) = a \cdot b_h(u, v).$$

Therefore,  $b_h$  is  $A$ -linear in the first argument and conjugate linear in the second.

Suppose that  $M$  is a right  $A$ -module. Since  $h$  is a right  $A$ -module homomorphism,  $h(ua) = h(u) \cdot a$  for all  $a \in A$  and  $u \in M$ . Thus,

$$b_h(ua, v) = h(ua)(v) = (h(u)a)(v) = \bar{a} \cdot h(u)(v)$$

for all  $u \in M$ . Moreover, for all  $a \in A$  and  $u \in M$ , since  $h(u) \in M^* = \text{Hom}_A(M, A)_\sigma$  is a right  $A$ -module homomorphism,

$$b_h(u, va) = h(u)(va) = (h(u)(v)) \cdot a = b_h(u, v) \cdot a.$$

Therefore,  $b_h$  is  $A$ -linear in the second argument and conjugate linear in the first.  $\square$

**Lemma 1.1.18** ([?, Chapter I, 2.2]). Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $M$  be a left/right  $A$ -module. A sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow A$  on  $M$  is determined by its adjoint (Definition 1.1.14)  $\hat{b} = h_b$ . Conversely,  $A$ -linear homomorphisms  $b : M \rightarrow M^*$  (Definition 1.1.11) are determined by their associated sesquilinear forms (Definition 1.1.16)  $b_h$ . In fact, the map  $b \mapsto b_h$  is a (two-sided)  $Z(A)$  (Definition 1.0.1)-module isomorphism

$$\text{Sesq}_A(M) \cong \text{Hom}_A(M, M^*).$$

**Corollary 1.1.19.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5).

1. Let  $M$  be a right  $A$ -module. There is a natural abelian group isomorphism

$$\text{Sesq}_A(M) \cong \text{Hom}_{A-A}(\overline{M} \otimes_{\mathbb{Z}} M, A)$$

(Definition 1.1.9)

2. Let  $M$  be a left  $A$ -module. There is a natural abelian group isomorphism

$$\text{Sesq}_A(M) \cong \text{Hom}_{A-A}(M \otimes_{\mathbb{Z}} \overline{M}, A)$$

(Definition 1.1.9)

*Proof.* 1. We use the isomorphism of Lemma 1.1.18 —  $\text{Sesq}_A(M)$  is naturally isomorphic as a  $Z(A)$ -bimodule to  $\text{Hom}_{\mathbb{Z}-A}(M, M^*)$ . Since  $\bar{\cdot}$  (Definition 1.1.9) is an involutive functor, this abelian group is naturally isomorphic to  $\text{Hom}_{A-\mathbb{Z}}(\overline{M}, M^\vee) = \text{Hom}_{A-\mathbb{Z}}(\overline{M}, \text{Hom}_{\mathbb{Z}-A}(M, A))$ . By the tensor-hom adjunction, this is naturally isomorphic to  $\text{Hom}_{A-A}(\overline{M} \otimes_{\mathbb{Z}} M, A)$  as desired.

2. This holds similarly.

$\square$

**Definition 1.1.20.** Let  $A$  be a ring with involution (Definition 1.0.5) denoted  $\bar{\cdot}$  and  $M$  a right  $A$ -module equipped with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow A$ .

Define a sesquilinear form  $b^* : M \times M \rightarrow A$ , which may be called by names such as the *conjugate transpose of  $b$*  or the *adjoint sesquilinear form of  $b$*  (not to be confused with the adjoint map  $M \rightarrow M^*$  of Definition 1.1.14), by

$$b^*(x, y) = \overline{b(y, x)}, \quad x, y \in M.$$

**Lemma 1.1.21** ([?, Chapter I, 2.3]). Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5) where  $\sigma$  is also denoted by  $\bar{\cdot}$ , let  $\varepsilon \in Z(A)$  be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ , let  $M$  be a right  $A$ -module, and let  $b : M \times M \rightarrow A$  be a sesquilinear form (Definition 1.1.1) on  $M$ .

We have

$$h_{b^*} = (h_b)^* \circ \text{can}_M$$

where  $b^*$  (Definition 1.1.20) is the conjugate transpose of  $b$ ,  $h_b$  (Definition 1.1.14) is the adjoint of  $b$ ,  $\text{can}_M$  (Definition 1.1.13) is the canonical map from  $M$  to its double twisted dual, and  $h_b^*$  is the map

$$h_b^* : M^{**} \rightarrow M^*$$

induced from  $h_b : M \rightarrow M^*$  by the contravariance of the twisted dual (Definition 1.1.11) functor  $(-)^*$ .

## 1.2. $\varepsilon$ -Hermitian forms on modules over involution rings.

**Definition 1.2.1.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5) where  $\sigma$  is also denoted by  $\bar{\cdot}$ , let  $\varepsilon \in Z(A)$  be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ , and let  $M$  be a (left or right)  $A$ -module. A  *$\varepsilon$ -hermitian form on  $M$*  (or simply *hermitian form on  $M$* , especially when  $\varepsilon$  is understood or is just 1) is a  $\sigma$ -sesquilinear form (Definition 1.1.1)  $\phi : M \times M \rightarrow A$  satisfying the symmetry condition:

$$\phi(y, x) = \varepsilon \cdot \overline{\phi(x, y)}$$

for all  $x, y \in M$ . The pair  $(M, \phi)$  is often called a *hermitian module*.

1-hermitian forms may simply be called *hermitian forms*, and  $(-1)$ -hermitian forms are called *skew-hermitian forms*.

As with sesquilinear forms (Definition 1.1.1) more generally,  $\varepsilon$ -Hermitian forms are either conjugate linear on the first variable and linear on the second variable or linear on the first variable and conjugate linear on the second variable, depending on whether  $M$  is regarded as a right  $A$ -module or a left  $A$ -module.

**Definition 1.2.2.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $\varepsilon \in Z(A)$  be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

The *category of modules with  $\varepsilon$ -Hermitian forms over  $(A, \sigma)$*  is the category whose objects are pairs  $(M, h)$  where  $M$  is a (right)  $A$ -module and  $h : M \times M \rightarrow A$  is a  $\varepsilon$ -Hermitian form (Definition 1.2.1) and whose morphisms are isometries (Definition 1.1.4).

In particular, such a category is a groupoid and is a full subcategory of the category of modules with sesquilinear forms over  $(A, \sigma)$  (Definition 1.1.5).

**Definition 1.2.3.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5) and  $M$  a right  $A$ -module equipped with a sesquilinear form (Definition 1.1.1)  $\phi : M \times M \rightarrow A$ . Let

$$\hat{\phi} : M \rightarrow M^*, \quad x \mapsto \phi(x, \cdot)$$

be the induced (left) adjoint map (Definition 1.1.14).

1. The form  $\phi$  is said to be **nondegenerate** if  $\hat{\phi}$  is injective.

In the case where  $\phi$  is an  $\varepsilon$ -hermitian form (Definition 1.2.1), this condition is equivalent to requiring that  $\phi(x, y) = 0$  for all  $y \in M$  implies  $x = 0$ .

2. The form  $\phi$  is said to be **nonsingular** or **regular** if  $\hat{\phi}$  is an isomorphism.

**Definition 1.2.4** ([?, Chapter I, 2.3]). Let  $A$  be a ring with involution (Definition 1.0.5) denoted by  $\bar{\cdot}$ , let  $\varepsilon \in Z(A)$  satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ , and let  $M$  be a (right)  $A$ -module.

The **form symmetrizer map** is the map

$$S_\varepsilon : \text{Sesq}_A(M) \rightarrow \text{Sesq}_A(M), \quad b \mapsto b + \varepsilon b^*,$$

(Definition 1.1.1) (Definition 1.1.20). It corresponds corresponds under the  $Z(A)$  (Definition 1.0.1)-module isomorphism

$$\begin{aligned} \text{Sesq}_A(M) &\cong \text{Hom}_A(M, M^*) \\ b &\mapsto h_b \\ b_h &\leftarrow h \end{aligned}$$

(Lemma 1.1.18) (Definition 1.1.14) (Definition 1.1.16) to the **operator symmetrizer map**

$$S_\varepsilon : \text{Hom}_A(M, M^*) \rightarrow \text{Hom}_A(M, M^*), \quad h \mapsto h + \varepsilon \cdot (h^* \circ \text{can}_M),$$

(Definition 1.1.13) where  $h^* : M^{**} \rightarrow M^*$  is the morphism induced from  $h : M \rightarrow M^*$  by the contravariance of the twisted dual (Definition 1.1.11) functor  $(-)^*$ . In other words, we have

$$h_{S_\varepsilon(b)} = S_\varepsilon(h_b)$$

(Definition 1.1.14).

**Notation 1.2.5** ([?, Chapter I, 2.3]). Let  $A$  be a ring with involution (Definition 1.0.5) denoted by  $\bar{\cdot}$ , let  $\varepsilon \in Z(A)$  satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ , and let  $M$  be a (right)  $A$ -module.

Write

$$\begin{aligned} \text{Sesq}_\varepsilon(M) &:= \text{Im}(S_\varepsilon) \subset \text{Sesq}_A(M) \\ \text{Sesq}^\varepsilon(M) &:= \text{Ker}(S_{-\varepsilon}) \subset \text{Sesq}_A(M). \end{aligned}$$

(Definition 1.2.4); in general,  $\text{Sesq}_\varepsilon(M) \subset \text{Sesq}^\varepsilon(M)$ .

The groups  $\text{Sesq}^\varepsilon(M)$  and  $\text{Sesq}_\varepsilon(M)$  are the groups of  $\varepsilon$ -Hermitian forms on  $M$  (Definition 1.2.1) and of even  $\varepsilon$ -Hermitian forms on  $M$  respectively.

### 1.3. Matrix representations of Hermitian forms on free modules.

**Definition 1.3.1.** (♠ TODO: AI generated, verify) Let  $R$  be a (not necessarily commutative) ring. Two matrices  $A, B \in M_n(R)$  are called **similar over  $R$**  (or **conjugate over  $R$** ) if there exists an invertible matrix  $P \in \text{GL}_n(R)$  such that

$$B = P^{-1}AP.$$

**Definition 1.3.2.** (♠ TODO: AI generated, verify) Let  $R$  be a (not necessarily commutative) ring and let  $M$  be a free (left/right)  $R$ -module of finite rank  $n$ .

Let  $\mathcal{E} = (e_1, \dots, e_n)$  and  $\mathcal{F} = (f_1, \dots, f_n)$  be two ordered bases of  $M$ . The *change-of-basis matrix from  $\mathcal{F}$  to  $\mathcal{E}$*  (or the *transition matrix from  $\mathcal{F}$  to  $\mathcal{E}$* ) is the matrix  $P \in M_n(R)$  whose  $j$ -th column consists of the coordinates of  $f_j$  in the basis  $\mathcal{E}$ :

$$P = (P(i, j))_{1 \leq i, j \leq n} \quad \text{where} \quad f_j = \sum_{i=1}^n e_i P(i, j)$$

if  $M$  is a right  $R$ -module, and

$$P = (P(i, j))_{1 \leq i, j \leq n} \quad \text{where} \quad f_j = \sum_{i=1}^n P(i, j) e_i$$

if  $M$  is a left  $R$ -module.

In either case,  $P$  is invertible and its inverse  $P^{-1}$  is the change-of-basis matrix from  $\mathcal{E}$  to  $\mathcal{F}$ . For any element  $x \in M$ , its coordinate column vectors  $[x]_{\mathcal{E}}$  and  $[x]_{\mathcal{F}}$  in the two bases satisfy

$$[x]_{\mathcal{E}} = P [x]_{\mathcal{F}}.$$

**Definition 1.3.3.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), where  $\sigma$  is denoted by  $a \mapsto \bar{a}$ .

For any matrix  $X \in M_{m \times n}(A)$ , the *conjugate transpose of  $X$*  (or *\*-adjoint of  $X$* ) is the  $n \times m$  matrix

$$X^* = \overline{X^T} = (\overline{X(j, i)})_{\substack{i=1, \dots, n \\ j=1, \dots, m}} \in M_{n \times m}(A).$$

The map  $X \mapsto X^*$  satisfies  $(X^*)^* = X$  and, when the dimensions are compatible,  $(XY)^* = Y^* X^*$  (Lemma 1.3.4). In the special case of square matrices,  $X \mapsto X^*$  is an involution (Definition 1.0.5) on the matrix ring  $M_n(A)$ , and this involution along with the embedding  $A \mapsto aI_n$  makes  $M_n(A)$  into an involution algebra (Definition 1.0.6) over  $A$ .

**Lemma 1.3.4.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), where  $\sigma$  is denoted by  $a \mapsto \bar{a}$ .

For any  $X \in M_{m \times n}(A)$  and  $Y \in M_{n \times p}(A)$ , we have

$$(XY)^* = Y^* X^*.$$

(Definition 1.3.3)

*Proof.* Write  $X = (x_{ij})$  and  $Y = (y_{jk})$  so that  $X$  is an  $m \times n$  matrix and  $Y$  is an  $n \times p$  matrix. The  $(i, k)$ -entry of  $XY$  is

$$(XY)_{ik} = \sum_{j=1}^n x_{ij} y_{jk}.$$

By Definition 1.3.3, the  $(k, i)$ -entry of  $(XY)^*$  is the conjugate of the  $(i, k)$ -entry of  $XY$ :

$$(XY)^*_{ki} = \overline{(XY)_{ik}} = \overline{\sum_{j=1}^n x_{ij} y_{jk}}.$$

Since  $\sigma$  is a ring involution, conjugation distributes over addition and reverses the order of multiplication:

$$\overline{\sum_{j=1}^n x_{ij} y_{jk}} = \sum_{j=1}^n \overline{x_{ij} y_{jk}} = \sum_{j=1}^n \overline{y_{jk}} \cdot \overline{x_{ij}}.$$

On the other hand,  $Y^*$  is a  $p \times n$  matrix with  $(k, j)$ -entry  $\overline{y_{jk}}$ , and  $X^*$  is an  $n \times m$  matrix with  $(j, i)$ -entry  $\overline{x_{ij}}$ . The  $(k, i)$ -entry of  $Y^* X^*$  is therefore

$$(Y^* X^*)_{ki} = \sum_{j=1}^n (Y^*)_{kj} (X^*)_{ji} = \sum_{j=1}^n \overline{y_{jk}} \cdot \overline{x_{ij}}.$$

The two expressions coincide entrywise, so  $(XY)^* = Y^* X^*$  as desired.  $\square$

**Definition 1.3.5.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $\varepsilon \in Z(A)$  satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

Let  $H \in M_n(A)$  be a matrix. We say that  $H$  is a  $\varepsilon$ -Hermitian matrix over  $(A, \sigma)$ : if

$$H^* = \bar{\varepsilon} H.$$

We remark that the choice of  $\bar{\varepsilon}$  here as opposed to  $\varepsilon$  is to ensure that the Hermitian form  $h_H$  associated to  $H$  (Definition 1.3.9) is a  $\varepsilon$ -Hermitian form (Definition 1.2.1).

When  $\varepsilon = 1$ ,  $H$  is simply called a *Hermitian matrix* (or *self-adjoint matrix*). When  $\varepsilon = -1$ ,  $H$  is called a *skew-Hermitian matrix*.

**Definition 1.3.6.** (♠ TODO: incorporate the choice of basis into the label) Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), where  $\sigma$  is denoted by  $a \mapsto \bar{a}$ . Let  $M$  be a free (left/right)  $A$ -module of finite rank  $n$  with ordered basis  $\mathcal{E} = (e_1, \dots, e_n)$ , and let  $h : M \times M \rightarrow A$  be an  $\varepsilon$ -Hermitian form (Definition 1.2.1) for some  $\varepsilon \in Z(A)$  (Definition 1.0.1) with  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

The *matrix of  $h$  with respect to  $\mathcal{E}$*  is the  $n \times n$  matrix

$$H_{\mathcal{E}} = (h(e_i, e_j))_{1 \leq i, j \leq n} \in M_n(A).$$

For any column vectors  $u, v \in A^n$  representing elements of  $M$  in the basis  $\mathcal{E}$ , the form evaluates in one of the following inequivalent ways:

$$h(x, y) = \begin{cases} u^* H_{\mathcal{E}} v & \text{if } M \text{ is a right } A\text{-module} \\ u^{\top} H_{\mathcal{E}} v & \text{if } M \text{ is a left } A\text{-module} \end{cases},$$

(Lemma 1.3.8)

The matrix  $H_{\mathcal{E}}$  is an  $\varepsilon$ -Hermitian matrix (Definition 1.3.5) (Lemma 1.3.7).

**Lemma 1.3.7.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), where  $\sigma$  is denoted by  $a \mapsto \bar{a}$ . Let  $M$  be a free (left/right)  $A$ -module of finite rank  $n$  with ordered basis  $\mathcal{E} = (e_1, \dots, e_n)$ , and let  $h : M \times M \rightarrow A$  be an  $\varepsilon$ -Hermitian form (Definition 1.2.1) for some  $\varepsilon \in Z(A)$  (Definition 1.0.1) with  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

(♠ **TODO: conjugate linear**) The matrix  $H_{\mathcal{E}} \in M_n(A)$  of  $h$  with respect to  $\mathcal{E}$  is indeed a  $\varepsilon$ -Hermitian matrix (Definition 1.3.5), i.e.  $H^* = \varepsilon H$  (Definition 1.3.3); this is true whether  $M$  is considered as left or a right module and hence whether  $H$  is conjugate linear in the first or the second variable.

*Proof.* We are to show that  $H_{\mathcal{E}}^* = \varepsilon H_{\mathcal{E}}$ . This is equivalent to say that that

$$\overline{h(e_j, e_i)} = \varepsilon \cdot h(e_i, e_j)$$

for all  $1 \leq i, j \leq n$ , which in turn is true because  $h$  is Hermitian.  $\square$

**Lemma 1.3.8.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), where  $\sigma$  is denoted by  $a \mapsto \bar{a}$ . Let  $M$  be a free (left/right)  $A$ -module of finite rank  $n$  with ordered basis  $\mathcal{E} = (e_1, \dots, e_n)$ , and let  $h : M \times M \rightarrow A$  be an  $\varepsilon$ -Hermitian form (Definition 1.2.1) for some  $\varepsilon \in Z(A)$  (Definition 1.0.1) with  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

1. Suppose that  $M$  is a right  $A$ -module (so that Hermitian forms are conjugate linear in the first variable and linear in the second (Definition 1.1.1))

For any column vectors  $u, v \in A^n$  representing elements of  $M$  in the basis  $\mathcal{E}$  (so  $x = \sum_i e_i u_i$ ,  $y = \sum_j e_j v_j$ ), we have

$$h(x, y) = u^* H_{\mathcal{E}} v,$$

(Definition 1.3.3)

2. Suppose that  $M$  is a left  $A$ -module (so that Hermitian forms are linear in the first variable and conjugate linear in the second (Definition 1.1.1))

For any column vectors  $u, v \in A^n$  representing elements of  $M$  in the basis  $\mathcal{E}$  (so  $x = \sum_i u_i e_i$ ,  $y = \sum_j v_j e_j$ ), we have

$$h(x, y) = u^{\top} H_{\mathcal{E}} \bar{v},$$

*Proof.* 1. We have

$$h(x, y) = h\left(\sum_i u_i e_i, \sum_j v_j e_j\right) = \sum_{i,j} \bar{u}_i \cdot h(e_i, e_j) \cdot v_j.$$

On the other hand,

$$\begin{aligned} u^* H_{\mathcal{E}} v &= (\bar{u}_1 \quad \cdots \quad \bar{u}_n) \cdot \begin{pmatrix} h(e_1, e_1) & \cdots & h(e_n, e_1) \\ \vdots & \ddots & \vdots \\ h(e_1, e_n) & \cdots & h(e_n, e_n) \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \\ &= \sum_{i,j} \bar{u}_i \cdot h(e_i, e_j) \cdot v_j \end{aligned}$$

so  $h(x, y) = u^* H_{\mathcal{E}} v$  indeed holds.

2. We have

$$h(x, y) = h\left(\sum_i u_i e_i, \sum_j v_j e_j\right) = \sum_{i,j} u_i \cdot h(e_i, e_j) \cdot \bar{v}_j.$$

On the other hand,

$$\begin{aligned} u^\top H_\mathcal{E} \bar{v} &= (u_1 \quad \cdots \quad u_n) \cdot \begin{pmatrix} h(e_1, e_1) & \cdots & h(e_n, e_1) \\ \vdots & \ddots & \vdots \\ h(e_1, e_n) & \cdots & h(e_n, e_n) \end{pmatrix} \cdot \begin{pmatrix} \bar{v}_1 \\ \vdots \\ \bar{v}_n \end{pmatrix} \\ &= \sum_{i,j} u_i \cdot h(e_i, e_j) \cdot \bar{v}_j \end{aligned}$$

so  $h(x, y) = u^\top H_\mathcal{E} v$  indeed holds.

□

**Definition 1.3.9.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $\varepsilon \in Z(A)$  (Definition 1.0.1) satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

Let  $H \in M_n(A)$  be an  $\varepsilon$ -Hermitian matrix (Definition 1.3.5). Let  $M = A^n$  be the free (left/right)  $A$ -module of column vectors with standard basis  $\mathcal{E} = (e_1, \dots, e_n)$ .

1. If  $M$  is a right  $A$ -module, then the  $\varepsilon$ -Hermitian form induced by  $H$  is the map

$$h_H : M \times M \rightarrow A, \quad h_H(u, v) = u^* H v,$$

where  $u, v \in A^n$  are column vectors. We note that  $h_H$  is conjugate linear in the first variable and linear in the second.

2. If  $M$  is a left  $A$ -module, then the  $\varepsilon$ -Hermitian form induced by  $H$  is the map

$$h_H : M \times M \rightarrow A, \quad h_H(u, v) = u^\top H \bar{v},$$

where  $u, v \in A^n$  are column vectors. We note that  $h_H$  is linear in the first variable and conjugate linear in the second.

The matrix of  $h_H$  with respect to  $\mathcal{E}$  is exactly  $H$  (♠ TODO: ). The form  $h_H$  is indeed an  $\varepsilon$ -Hermitian form (Definition 1.2.1) on  $M$  (Lemma 1.3.10).

**Lemma 1.3.10.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $\varepsilon \in Z(A)$  (Definition 1.0.1) satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ . Let  $H \in M_n(A)$  be an  $\varepsilon$ -Hermitian matrix (Definition 1.3.5) in the sense that

$$H = \bar{\varepsilon} \cdot H^*.$$

Let  $M = A^n$  be the free (left/right)  $A$ -module of column vectors with standard basis  $\mathcal{E} = (e_1, \dots, e_n)$ .

The  $\varepsilon$ -Hermitian form  $h_H$  induced by  $H$  (Definition 1.3.9) is indeed a  $\varepsilon$ -Hermitian form (Definition 1.2.1) on  $M$ ; in particular,

$$h_H(u, v) = \varepsilon \cdot \overline{h_H(v, u)}$$

for all  $u, v \in M$ .

*Proof.* Suppose that  $M$  is a right  $A$ -module and hence  $h_H$  is defined as

$$h_H : M \times M \rightarrow A, \quad h_H(u, v) = u^* H v.$$

We show that  $h_H$  is conjugate linear in the first variable and linear in the second — given  $a, b \in A$ , we have

$$h_H(ua, vb) = (ua)^* H(vb) = \bar{a} \cdot u^* H v \cdot b = \bar{a} \cdot h_H(u, v) \cdot b.$$

Moreover, assuming that  $H$  is  $\varepsilon$ -Hermitian in the sense that

$$H^* = \bar{\varepsilon} H,$$

we have

$$h_H(v, u) = v^* H u = (u^* H^* v)^*.$$

By Lemma 1.3.4,  $(u^* H^* v)^* = v^* H u$ , and since this is a  $1 \times 1$  matrix, it equals its own transpose, so

$$\overline{u^* H^* v} = \overline{u^* \cdot (\bar{\varepsilon} H) \cdot v} = \varepsilon \cdot \overline{u^* H v} = \varepsilon \cdot \overline{h_H(u, v)}$$

Similarly, if  $M$  is a left  $A$ -module, then  $h_H$  is linear in the first variable and conjugate linear in the second. Moreover, assuming that  $H$  is  $\varepsilon$ -Hermitian in the sense that

$$H^* = \varepsilon H,$$

we have

$$h_H(v, u) = v^\top H \bar{u} = (u^\top H^* \bar{v})^*.$$

As this is a  $1 \times 1$  matrix, it equals its own transpose

$$\overline{u^\top H^* \bar{v}} = \overline{u^\top \cdot (\varepsilon H) \cdot \bar{v}} = \varepsilon \cdot \overline{u^\top H \bar{v}} = \varepsilon \cdot \overline{h_H(u, v)}$$

□

**Definition 1.3.11.** (♠ TODO: AI generated, verify) Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5).

Two matrices  $H, K \in M_n(A)$  are called *\*-congruent* (or *conjugate-transpose congruent*) if there exists an invertible matrix  $P \in \text{GL}_n(A)$  such that

$$K = P^* H P,$$

where  $P^*$  denotes the conjugate transpose of  $P$  (Definition 1.3.3).

If  $H$  is an  $\varepsilon$ -Hermitian matrix (Definition 1.3.5) and  $K$  is \*-congruent to  $H$ , then  $K$  is also  $\varepsilon$ -Hermitian (Lemma 1.3.12).

Let  $M$  be a free (right)  $A$ -module of rank  $n$  with an  $\varepsilon$ -Hermitian form (Definition 1.2.1)  $h$ . If  $\mathcal{E}$  and  $\mathcal{F}$  are two ordered bases of  $M$  with change-of-basis matrix  $P \in \text{GL}_n(A)$  (so  $f_j = \sum_i e_i P(i, j)$ ), then

$$H_{\mathcal{F}} = P^* H_{\mathcal{E}} P,$$

where  $H_{\mathcal{E}}$  and  $H_{\mathcal{F}}$  are the matrices of  $h$  with respect to  $\mathcal{E}$  and  $\mathcal{F}$  (Definition 1.3.6), respectively (Lemma 1.3.13). Thus the two matrices are \*-congruent.

**Lemma 1.3.12.** Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5).

Let  $H, K \in M_n(A)$  be conjugate-transpose congruent (Definition 1.3.11). If  $H$  is  $\varepsilon$ -Hermitian (Definition 1.3.5), then so is  $K$ .

*Proof.* Say that  $P \in \text{GL}_n(A)$  satisfies  $K = P^*HP$ . We have that  $H^* = \bar{\varepsilon}H$ , so

$$K^* = (P^*HP)^* = P^*H^*P = P^*(\bar{\varepsilon}H)P = \bar{\varepsilon}(P^*HP) = \bar{\varepsilon}K.$$

□

**Lemma 1.3.13.** (♠ TODO: AI generated, verify) Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), where  $\sigma$  is denoted by  $a \mapsto \bar{a}$ . Let  $M$  be a free (left/right)  $A$ -module of finite rank  $n$  with an  $\varepsilon$ -Hermitian form (Definition 1.2.1)  $h : M \times M \rightarrow A$  for some  $\varepsilon \in Z(A)$  (Definition 1.0.1) with  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

Let  $\mathcal{E} = (e_1, \dots, e_n)$  and  $\mathcal{F} = (f_1, \dots, f_n)$  be two ordered bases of  $M$ , and let  $P \in \text{GL}_n(A)$  be the change-of-basis matrix from  $\mathcal{F}$  to  $\mathcal{E}$  (Definition 1.3.2). Denote by  $H_{\mathcal{E}}$  and  $H_{\mathcal{F}}$  the matrices of  $h$  with respect to  $\mathcal{E}$  and  $\mathcal{F}$  (Definition 1.3.6), respectively.

1. If  $M$  is a right  $A$ -module (so that  $h$  is conjugate linear in the first variable and linear in the second (Definition 1.1.1)), then

$$H_{\mathcal{F}} = P^* H_{\mathcal{E}} P,$$

where  $P^*$  denotes the conjugate transpose of  $P$  (Definition 1.3.3).

2. If  $M$  is a left  $A$ -module (so that  $h$  is linear in the first variable and conjugate linear in the second (Definition 1.1.1)), then

$$H_{\mathcal{F}} = P^{\top} H_{\mathcal{E}} \bar{P},$$

where  $P^{\top}$  denotes the transpose of  $P$  and  $\bar{P}$  denotes the entrywise conjugate of  $P$ .

In particular,  $H_{\mathcal{E}}$  and  $H_{\mathcal{F}}$  are  $*$ -congruent (Definition 1.3.11) when  $M$  is a right  $A$ -module. For left  $A$ -modules, the relation  $H_{\mathcal{F}} = P^{\top} H_{\mathcal{E}} \bar{P}$  is not  $*$ -congruence in general, but reduces to  $*$ -congruence when the involution is trivial (e.g. when  $A$  is a commutative ring with the identity involution).

*Proof.* (♠ TODO: AI generated, verify)

1. Suppose  $M$  is a right  $A$ -module. For any  $x, y \in M$ , write  $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$  and  $[y]_{\mathcal{E}} = P[y]_{\mathcal{F}}$ . By Lemma 1.3.8, we have

$$\begin{aligned} h(x, y) &= [x]_{\mathcal{E}}^* H_{\mathcal{E}} [y]_{\mathcal{E}} \\ &= (P[x]_{\mathcal{F}})^* H_{\mathcal{E}} (P[y]_{\mathcal{F}}) \\ &= [x]_{\mathcal{F}}^* (P^* H_{\mathcal{E}} P) [y]_{\mathcal{F}}. \end{aligned}$$

Since  $h(x, y) = [x]_{\mathcal{F}}^* H_{\mathcal{F}} [y]_{\mathcal{F}}$  by the same lemma, and this holds for all  $x, y \in M$ , we conclude  $H_{\mathcal{F}} = P^* H_{\mathcal{E}} P$ .

2. Suppose  $M$  is a left  $A$ -module. For any  $x, y \in M$ , write  $[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}$  and  $[y]_{\mathcal{E}} = P[y]_{\mathcal{F}}$ . By Lemma 1.3.8, we have

$$\begin{aligned} h(x, y) &= [x]_{\mathcal{E}}^{\top} H_{\mathcal{E}} \overline{[y]_{\mathcal{E}}} \\ &= (P[x]_{\mathcal{F}})^{\top} H_{\mathcal{E}} \overline{(P[y]_{\mathcal{F}})} \\ &= [x]_{\mathcal{F}}^{\top} (P^{\top} H_{\mathcal{E}} \bar{P}) \overline{[y]_{\mathcal{F}}}. \end{aligned}$$

Since  $h(x, y) = [x]_{\mathcal{F}}^{\top} H_{\mathcal{F}} \overline{[y]_{\mathcal{F}}}$  by the same lemma, and this holds for all  $x, y \in M$ , we conclude  $H_{\mathcal{F}} = P^{\top} H_{\mathcal{E}} \overline{P}$ .

□

**Definition 1.3.14.** (♠ TODO: AI generated, verify) Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $H \in M_n(A)$  be an  $\varepsilon$ -Hermitian matrix (Definition 1.3.5) for some  $\varepsilon \in Z(A)$  with  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

An invertible matrix  $U \in \text{GL}_n(A)$  is called a *unitary transformation for  $H$*  (or simply  *$H$ -unitary*) if

$$U^* H U = H.$$

Equivalently,  $U$  is unitary for  $H$  if and only if  $U$  preserves  $h_H$ .

The set of all such matrices forms a group

$$U(H) = \{U \in \text{GL}_n(A) : U^* H U = H\},$$

called the *unitary group of  $H$* . It is the subgroup of  $\text{GL}_n(A)$  that fixes  $H$  under the  $*$ -congruence (Definition 1.3.11) action.

#### 1.4. Sesquilinear and Hermitian modules and spaces.

**Definition 1.4.1** ([?, Chapter I, 2.3]). Let  $A$  be a ring with involution (Definition 1.0.5), let  $\varepsilon \in Z(A)$  satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ , and let  $M$  be an (right)  $A$ -module. Write  $\text{Sesq}_{\varepsilon}(M)$  and  $\text{Sesq}^{\varepsilon}(M)$  as in Notation 1.2.5.

One calls elements of  $\text{Sesq}_{\varepsilon}(M)$  *even  $\varepsilon$ -Hermitian forms*; in other words, a sesquilinear form (Definition 1.1.1)  $b \in \text{Sesq}_A(M)$  is an even  $\varepsilon$ -Hermitian form if and only if there exists some sesquilinear form  $k \in \text{Sesq}_A(M)$  such that  $b = k + \varepsilon k^*$  (Definition 1.1.14).

In general,  $\text{Sesq}_{\varepsilon}(M) \subset \text{Sesq}^{\varepsilon}(M)$ . If 2 is a unit in  $A$ , then  $\text{Sesq}_{\varepsilon}(M) = \text{Sesq}^{\varepsilon}(M)$ , since  $b = \frac{1}{2} S_{\varepsilon}(b)$  for  $b \in \text{Sesq}^{\varepsilon}(M)$ .

##### 1.4.1. Sesquilinear and Hermitian modules.

#### 1.5. Sesquilinear and Hermitian modules and spaces.

##### 1.5.1. Sesquilinear and Hermitian modules.

**Definition 1.5.1.** Let  $A$  be a ring with involution (Definition 1.0.5).

A *sesquilinear module over  $A$*  is a pair  $(M, b)$  with  $M$  a finitely generated projective (right)  $A$ -module and  $b$  a sesquilinear form (Definition 1.1.1) on  $M$ .

We may identify  $b$  with its adjoint (Definition 1.1.14) and write  $(M, h)$  to denote an  $\varepsilon$ -Hermitian module with  $h \in \text{Hom}_A(M, M^*)$  (Definition 1.1.11).

**Remark 1.5.2.** This terminology is not strictly present in [?].

**Definition 1.5.3** ([?, Chapter I, 3.1]). Let  $A$  be a ring with involution (Definition 1.0.5), and let  $\varepsilon \in Z(A)$  satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

1. A  $\varepsilon$ -Hermitian module over  $A$  is a sesquilinear module (Definition 1.5.1)  $(M, b)$  over  $A$  such that the sesquilinear form (Definition 1.1.1)  $b$  is in fact a  $\varepsilon$ -Hermitian form (Definition 1.2.1) on  $M$ .

In other words, a  $\varepsilon$ -Hermitian module over  $A$  is a pair  $(M, b)$  with  $M$  a finitely generated projective (right)  $A$ -module and  $b$  a  $\varepsilon$ -Hermitian form (Definition 1.2.1) on  $M$ .

As with sesquilinear modules, we may identify  $b$  with its adjoint (Definition 1.1.14) and write  $(M, h)$  to denote an  $\varepsilon$ -Hermitian module with  $h \in \text{Hom}_A(M, M^*)$  (Definition 1.1.11).

2. An *even  $\varepsilon$ -hermitian module over  $A$*  is a  $\varepsilon$ -hermitian module whose underlying  $\varepsilon$ -hermitian form is even (Definition 1.4.1).
3. A *symmetric bilinear form module over  $A$*  is an 1-Hermitian module over  $A$  where  $A$  has the trivial involution.
4. A *skew-symmetric bilinear form module over  $A$*  is a  $(-1)$ -Hermitian module over  $A$  where  $A$  has the trivial involution.

**Definition 1.5.4** (cf. [?, Chapter I, 3.2]). Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $\varepsilon \in Z(A)$  be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

The *category of  $\varepsilon$ -Hermitian modules over  $(A, \sigma)$*  is the full subcategory  $\mathfrak{Herm}^\varepsilon(A)$  of the category of modules with  $\varepsilon$ -hermitian forms over  $(A, \sigma)$  (Definition 1.2.2) whose objects are the  $\varepsilon$ -Hermitian modules (Definition 1.5.3), i.e. pairs  $(M, h)$  where  $M$  is a finitely generated projective (one-sided)  $A$ -module.

The *category of even  $\varepsilon$ -Hermitian modules over  $(A, \sigma)$*  is the full subcategory  $\mathfrak{Herm}_+^\varepsilon(A)$  of  $\mathfrak{Herm}^\varepsilon(A)$  whose objects are the even (Definition 1.4.1)  $\varepsilon$ -hermitian modules over  $A$ .

In particular, such categories are groupoids.

### 1.5.2. Orthogonality in sesquilinear and Hermitian modules.

**Definition 1.5.5.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be a (left/right) module over  $R$  with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow R$ .

We say that an element  $x \neq 0$  of  $M$  is *isotropic for  $(M, b)$*  if  $b(x, x) = 0$ .

**Definition 1.5.6.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be a (left/right) module over  $R$  with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow R$ .

We say that  $(M, b)$  is *isotropic* if there is a nonzero element  $x \in M$  that is isotropic (Definition 1.5.5).

**Definition 1.5.7.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be a (left/right) module over  $R$  with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow R$ .

We say that  $(M, b)$  is *anisotropic* if  $b(x, x) = 0$  for  $x \in M$  implies  $x = 0$ .

**Definition 1.5.8.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be a (left/right) module over  $R$  with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow R$ .

We say that two subsets  $X, Y \subset M$  are *orthogonal* if  $b(x, y) = 0$  for all  $x \in X$  and  $y \in Y$ .

**Definition 1.5.9.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be a (left/right) module over  $R$  with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow R$ .

We say that a submodule  $U$  of  $M$  is *totally isotropic (for  $(M, b)$ )* if  $b|_U = 0$ .

1. We say that  $U$  is *totally left isotropic (for  $(M, b)$ )* if  $b(u, m) = 0$  for all  $u \in U$  and  $m \in M$ .
2. We say that  $U$  is *totally right isotropic (for  $(M, b)$ )* if  $b(m, u) = 0$  for all  $u \in U$  and  $m \in M$ .
3. We say that  $U$  is *totally isotropic (for  $(M, b)$ )* if  $b(u, m) = 0$  and  $b(m, u) = 0$  for all  $u \in U$  and  $m \in M$ .

Letting  $\varepsilon \in Z(R)$  (Definition 1.0.1) be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$  and assuming that  $b$  is a  $\varepsilon$ -Hermitian form, to say that  $U$  is totally isotropic is  $(M, b)$  is equivalent to saying that  $U \subseteq U^\perp$ .

**Definition 1.5.10.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be a (left/right) module over  $R$  with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow R$ .

Let  $U \subset M$  be a submodule.

1. The *left orthogonal complement of  $U$  in  $(M, b)$*  is

$$U^{\perp_L} = \{x \in M : b(x, u) = 0 \text{ for all } u \in U\}.$$

2. The *right orthogonal complement of  $U$  in  $(M, b)$*  is

$$U^{\perp_R} = \{y \in M : b(u, y) = 0 \text{ for all } u \in U\}.$$

3. Let  $\varepsilon \in Z(R)$  (Definition 1.0.1) be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$  and Assume that  $b$  is a  $\varepsilon$ -Hermitian form. The left and right orthogonal complements of  $U$  in  $(M, b)$  coincide (Lemma 1.5.12); it is called the *orthogonal complement of  $U$  in  $(M, b)$*  and may be denoted by  $U^\perp$ .

**Lemma 1.5.11.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $(M, b)$  be a (left/right) module over  $R$  with a sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow R$ . Let  $U \subset M$  be a submodule.

1. The left orthogonal complement (Definition 1.5.10)  $U^{\perp_L}$  is a submodule of  $M$ .
2. The right orthogonal complement (Definition 1.5.10)  $U^{\perp_R}$  is a submodule of  $M$ .

Consequently, letting  $\varepsilon \in Z(R)$  (Definition 1.0.1) be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$  and assuming that  $b$  is a  $\varepsilon$ -Hermitian form, the orthogonal complement  $U^\perp$  is a submodule of  $M$ . (♠ TODO: Do a thing that says that it suffices to show nonemptiness, closure under addition and scalar multiplication to show that a subset is a submodule)

*Proof.* 1. Clearly,  $U^{\perp_L}$  contains 0. Moreover, for all  $x_1, x_2 \in U^{\perp_L}$ , we have

$$b(x_1 + x_2, u) = 0$$

for all  $u \in U$ , so  $U^{\perp_L}$  is closed under addition. Moreover, for all  $x \in M$  and  $r \in R$ ,

- Either  $M$  is a right  $R$ -module in which case

$$b(xr, u) = \bar{r} \cdot b(x, u) = 0,$$

or

- $M$  is a left  $R$ -module in which case

$$b(rx, u) = r \cdot b(x, u) = 0.$$

Thus,  $U$  is an  $R$ -submodule of  $M$ .

2. This holds similarly. □

**Lemma 1.5.12.** Let  $R$  be a ring with involution (Definition 1.0.5). Let  $\varepsilon \in Z(R)$  (Definition 1.0.1) be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ . Let  $(M, b)$  be a (left/right) module over  $R$  with a Hermitian form (Definition 1.2.1)  $b : M \times M \rightarrow R$ .

Let  $U \subset M$  be a submodule. The left and right orthogonal complements (Definition 1.5.10) of  $U$ , i.e. the sets

$$U^{\perp_L} = \{x \in M : b(x, u) = 0 \text{ for all } u \in U\}$$

and

$$U^{\perp_R} = \{y \in M : b(u, y) = 0 \text{ for all } u \in U\},$$

coincide. Thus, the orthogonal complement (Definition 1.5.10)  $U^\perp$  is well defined.

*Proof.* Suppose that  $x \in M$  satisfies  $b(x, u) = 0$  for all  $u \in U$ . Note that  $b(x, u) = \varepsilon \cdot \overline{b(u, x)}$ , so  $b(u, x) = 0$  for all  $u \in U$ . Thus,  $U^{\perp_L} \subseteq U^{\perp_R}$ . Similarly, the other inclusion holds, so the two sets are equal. □

### 1.5.3. Sesquilinear and Hermitian spaces.

**Definition 1.5.13** (cf. [?, Chapter I, 3.2]). Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5).

1. A **sesquilinear space** is a sesquilinear module (Definition 1.5.1)  $(M, b)$  whose underlying sesquilinear form  $b$  is nonsingular (Definition 1.2.3).
2. A  **$\varepsilon$ -hermitian space** is a  $\varepsilon$ -hermitian module (Definition 1.5.3) whose underlying Hermitian form is nonsingular (Definition 1.2.3).
3. An **even  $\varepsilon$ -hermitian space** is a  $\varepsilon$ -hermitian space (Definition 1.5.3) whose underlying Hermitian form is even (Definition 1.4.1).

In each case, note that a “space” is simply a “module” whose underlying sesquilinear form is nonsingular.

**Definition 1.5.14** (cf. [?, Chapter I, 3.2]). Let  $(A, \sigma)$  be a ring with involution (Definition 1.0.5), and let  $\varepsilon \in Z(A)$  be an element such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

The *category of  $\varepsilon$ -Hermitian spaces over  $(A, \sigma)$*  is the full subcategory  $\mathfrak{H}^\varepsilon(A)$  of the category of  $\varepsilon$ -hermitian spaces over  $(A, \sigma)$  (Definition 1.2.2) whose objects are the  $\varepsilon$ -Hermitian spaces (Definition 1.5.3), i.e. pairs  $(M, h)$  where  $M$  is a finitely generated projective (one-sided)  $A$ -module and  $h$  is nonsingular (Definition 1.2.3).

The *category of even  $\varepsilon$ -Hermitian spaces over  $(A, \sigma)$*  is the full subcategory  $\mathfrak{H}_+^\varepsilon(A)$  of  $\mathfrak{H}^\varepsilon(A)$  whose objects are the even (Definition 1.4.1)  $\varepsilon$ -hermitian spaces over  $A$ .

In particular, such categories are groupoids.

**Proposition 1.5.15.** Let  $(R, \sigma)$  be a ring with involution (Definition 1.0.5) and  $\phi : M \times M \rightarrow R$  a sesquilinear form (Definition 1.1.1). If  $M$  is a finitely generated projective  $R$ -module, then the adjoint map  $\hat{\phi} : M \rightarrow \text{Hom}_R(M, R)_\sigma$  is an isomorphism if and only if  $\phi$  is non-degenerate (Definition 1.2.3).

## 1.6. Unitary rings and unitary modules over unitary rings.

**Definition 1.6.1** (see [?, Chapter I, 5.1]). Let  $A$  be an involution ring (Definition 1.0.5). Let  $\varepsilon \in Z(A)$  (Definition 1.0.1) such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ . Let  $\Lambda$  be an additive subgroup of  $A$  such that

(♠ TODO:  $S_{-\varepsilon}(A), S^{-\varepsilon}(A)$ )

1.  $S_{-\varepsilon}(A) \subseteq \Lambda \subseteq S^{-\varepsilon}(A)$
2.  $a \cdot x \cdot \bar{a} \in \Lambda$  for all  $x \in \Lambda$  and  $a \in A$ .

The triple  $(A, \varepsilon, \Lambda)$  is called a *unitary ring* and the pair  $(\varepsilon, \Lambda)$  is called a form parameter for  $A$ .

A *morphism of unitary rings*  $f : (A, \varepsilon, \Lambda) \rightarrow (A', \varepsilon', \Lambda')$  is a ring homomorphism  $f : A \rightarrow A'$  such that  $f(\varepsilon) = \varepsilon'$  and  $f(\Lambda) \subset \Lambda'$ .

**Definition 1.6.2** (see [?, Chapter I, 5.1]). Let  $R$  be an involution ring (Definition 1.0.5). A *unitary  $R$ -algebra* is a unitary ring (Definition 1.6.1)  $(A, \varepsilon, \Lambda)$  such that  $A$  is an  $R$ -algebra with an  $R$ -linear involution (Definition 1.0.8) and  $\Lambda$  is an  $R$ -submodule of  $A$  such that  $S_{-\varepsilon}(A) \subseteq \lambda \subseteq S^{-\varepsilon}(A)$  and  $a \cdot x \cdot \bar{a} \in \Lambda$  for all  $x \in \Lambda$  and  $a \in \lambda$ .

(♠ TODO:  $S_{-\varepsilon}(A), S^{-\varepsilon}(A)$ )

**Lemma 1.6.3** (see [?, Chapter I, 5.1]). Let  $(R, \varepsilon, \Lambda)$  be a commutative unitary ring (Definition 1.6.1).

For any commutative  $R$ -algebra  $R'$ , the triple  $(A \otimes_R R', \varepsilon' = \varepsilon \otimes 1, \Lambda' = \Lambda \otimes_R R')$  is a unitary  $R'$ -algebra (Definition 1.6.2).

**Definition 1.6.4.** (♠ TODO: Sesq Lambda epsilon)

## 1.7. Base change of Sesquilinear forms, Hermitian forms, and Unitary modules.

**Lemma 1.7.1** ([?, Chapter I, 7.1]). Let  $\eta \rightarrow A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Given a right  $A$ -module  $M$  and there is a right  $A'$ -module homomorphism

$$j_M : M^* \otimes_A A' \rightarrow (M \otimes_A A')^* = \text{Hom}_{A'}(M', A')$$

(Definition 1.1.11) (??) (??) given by

$$j_M(f \otimes a')(m \otimes b') = \bar{a}'\eta(f(m))b'$$

for  $f \in M^*$ , and  $a', b' \in A'$ . The maps  $j_M$  specify a natural transformation in the category of right  $A'$ -modules.

If  $M$  is a projective right  $A$ -module, then  $j_M$  is an isomorphism.

**Definition 1.7.2** ([?, Chapter I, 7.1]). Let  $\eta \rightarrow A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Let  $M$  be a right  $A$ -module and let  $b : M \times M \rightarrow A$  be a sesquilinear form (Definition 1.1.1) on  $M$ .

By the *base change of  $b$*  (which may also be called by names such as *extension of scalars of  $b$*  or the *induction of  $b$  along  $\eta$*  as per ??) as the sesquilinear form  $b \otimes_A A' : (M \otimes_A A') \times (M \otimes_A A') \rightarrow A'$  (denoted by  $b'$  or  $b \otimes 1_{A'}$  in [?]) on the right  $A'$ -module  $M \otimes_A A'$  given by

$$(b \otimes_A A')(m \otimes a', n \otimes a'') = \bar{a}'\eta(b(m, n))a'' \quad \text{for all } m, n \in M, a', a'' \in A'.$$

and extended  $A'$ -sesquilinearly.

The pair  $(M \otimes_A A', b \otimes_A A')$  may also be written as  $(M, b) \otimes_A A'$ . (♠ TODO: discuss adjoint of base change)

**Lemma 1.7.3.** Let  $R$  and  $S$  be rings (not necessarily commutative), and let  $f : R \rightarrow S$  be a ring homomorphism. Let  $M$  be a projective right (resp. left)  $R$ -module. The base change  $M \otimes_R S$  (resp.  $S \otimes_R M$ ) is a projective right (resp. left)  $S$ -module.

*Proof.* A module over a ring is projective if and only if it is a direct summand of a free module. In particular  $M$  is a direct summand of a free  $R$ -module, so  $M \otimes_R S$  is a direct summand of a free  $S$ -module and hence is a projective  $S$ -module.  $\square$

**Lemma 1.7.4.** Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Let  $M$  be a right  $A$ -module and let  $b : M \times M \rightarrow A$  be a sesquilinear form (Definition 1.1.1) on  $M$ .

1. If  $(M, b)$  is a sesquilinear module (Definition 1.5.1) over  $A$ , then  $(M, b) \otimes_A A'$  is a sesquilinear module over  $A'$ .
2. Let  $\varepsilon \in Z(A)$  satisfy  $\varepsilon \cdot \bar{\varepsilon} = 1$ . If  $b$  is a  $\varepsilon$ -hermitian form (Definition 1.2.1) over  $A$ , then  $b \otimes_A A'$  is an  $\eta(\varepsilon)$ -hermitian form (Definition 1.2.1) over  $A'$ . Moreover, if  $b$  is even (Definition 1.4.1), then so is  $b \otimes_A A'$ .

*Proof.* 1. Since a sesquilinear module is simply a projective module equipped with a sesquilinear form, this follows from Lemma 1.7.3

2. To show that  $b \otimes_A A'$  is an  $\eta(\varepsilon)$ -hermitian form, it suffices to verify

$$(b \otimes_A A')(m \otimes a', n \otimes a'') = \eta(\varepsilon) \overline{(b \otimes_A A')(n \otimes a'', m \otimes a')}$$

for all  $m, n \in M$  and  $a', a'' \in A'$ . The left equals  $\overline{a'}\eta(b(m, n))a''$ , whereas the right equals

$$\begin{aligned} \eta(\varepsilon) \overline{\overline{a''}\eta(b(n, m))a'} &= \eta(\varepsilon) \overline{\overline{a'}\eta(b(n, m))\overline{a''}} \\ &= \eta(\varepsilon) \overline{a'}\eta(\overline{b(n, m)})\overline{a''} \\ &= \eta(\varepsilon) \overline{a'}\eta(\overline{\varepsilon \overline{b(m, n)}})\overline{a''} \\ &= \eta(\varepsilon) \overline{a'}\eta(\overline{b(m, n)} \cdot \overline{\varepsilon})\overline{a''} \\ &= \eta(\varepsilon) \overline{a'}\eta(b(m, n)\overline{\varepsilon})\overline{a''} \\ &= \overline{a'}\eta(\varepsilon \overline{\varepsilon} b(m, n))\overline{a''} \\ &= \overline{a'}\eta(b(m, n))\overline{a''} \end{aligned}$$

(♠ TODO: prove evenness)

□

### 1.8. Transfer of sesquilinear forms via a morphism of involution rings.

**Definition 1.8.1** (see [?, Chapter I, 7.2]). Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Given a right  $A'$ -module  $M'$ , write  $M'_{(A)}$  for the restriction of  $M'$  as a right  $A$ -module (via  $\eta$ ). Let  $\tau : A' \rightarrow A$  be a morphism of  $A$ -bimodules.

$\tau$  induces an additive map

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.1.1) that sends a sesquilinear form  $b' \in \text{Sesq}_{A'}(M')$  on an  $A'$ -module  $M'$  to the sesquilinear form  $\tau(b') \in \text{Sesq}_A(M'_{(A)})$  on  $M'_{(A)}$  constructed by

$$\tau(b')(x, y) = \tau(b'(x, y)).$$

It may be appropriate to call the induced map  $\tau$  the *transfer* or *Scharlau transfer on sesquilinear forms*.

**Lemma 1.8.2.** Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Given a (right)  $A'$ -module  $M'$ , write  $M'_{(A)}$  for the restriction of  $M'$  as a right  $A$ -module (via  $\eta$ ). Let  $\tau : A' \rightarrow A$  be a morphism of  $A$ -bimodules.

The Scharlau transfer

$$b' \mapsto (\tau(b') : (x, y) \mapsto \tau(b'(x, y)))$$

is indeed an additive map

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)}).$$

(Definition 1.1.1)

*Proof.* For all  $a_1, a_2 \in A$  and  $m, n \in M'$ , we have

$$\tau(b')(ma_1, na_2) = \tau(b'(ma_1, na_2)) = \tau(\bar{a}_1 b'(m, n) a_2) = \bar{a}_1 \tau(b'(m, n)) a_2 = \bar{a}_1 \cdot \tau(b')(m, n) \cdot a_2,$$

Therefore,  $\tau(b')$  is conjugate linear in the first argument and linear in the second.  $\square$

**Lemma 1.8.3** (see [?, Chapter I, 7.2]). Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Given a right  $A'$ -module  $M'$ , write  $M'_{(A)}$  for the restriction of  $M'$  as a right  $A$ -module (via  $\eta$ ). Let  $\tau : A' \rightarrow A$  be a morphism of  $A$ -bimodules and assume that

$$\tau(\bar{a}) = \overline{\tau(a)}$$

for all  $a \in A'$

The Scharlau transfer (Definition 1.8.1)

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.1.1) restricts to maps

$$\tau : \text{Sesq}^{\varepsilon'}(M') \rightarrow \text{Sesq}^{\varepsilon}(M'_{(A)})$$

$$\tau : \text{Sesq}_{-\varepsilon'}(M') \rightarrow \text{Sesq}_{-\varepsilon}(M'_{(A)})$$

(Notation 1.2.5)

**Definition 1.8.4** (see [?, Chapter I, 7.2]). Let  $\eta : (A, \Lambda, \varepsilon) \rightarrow (A', \Lambda', \varepsilon')$  be a homomorphism of unitary rings (Definition 1.6.1). Write  $\bar{\cdot}$  for both of the involutions on  $A$  and  $A'$ . Let  $\tau : A' \rightarrow A$  be a morphism of  $A$ -bimodules and assume that

$$\tau(\bar{a}) = \overline{\tau(a)}$$

for all  $a \in A'$ .

The Scharlau transfer (Definition 1.8.1)

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.1.1) restricts to a map

$$\tau : \text{Sesq}_{\Lambda'}^{\varepsilon'}(M') \rightarrow \text{Sesq}_{\Lambda}^{\varepsilon}(M'_{(A)})$$

**(♠ TODO: Sesq)** It is appropriate to also call this induced  $\tau$  the *Scharlau transfer for sesquilinear forms on modules over unitary rings*

**Definition 1.8.5** (see [?, Chapter I, 7.2]). Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6), and assume that  $A'$  is finitely generated and projective as a right  $A$ -module. Let  $\varepsilon \in A$  be an element satisfying  $\varepsilon \cdot \bar{\varepsilon} = 1$  and let  $\varepsilon' = \eta(\varepsilon)$ . Given a right  $A'$ -module  $M'$ , write  $M'_{(A)}$  for the restriction of  $M'$  as a right  $A$ -module (via  $\eta$ ). Let  $\tau : A' \rightarrow A$  be a morphism of  $A$ -bimodules and assume that

$$\tau(\bar{a}) = \overline{\tau(a)}$$

for all  $a \in A'$ .

The Scharlau transfer (Definition 1.8.4) map

$$\tau : \text{Sesq}_{A'}(M') \rightarrow \text{Sesq}_A(M'_{(A)})$$

(Definition 1.8.1) induces functors

$$\begin{aligned} T : \mathfrak{Herm}^{\varepsilon'}(A') &\rightarrow \mathfrak{Herm}^{\varepsilon}(A) \\ T : \mathfrak{Herm}_+^{\varepsilon'}(A') &\rightarrow \mathfrak{Herm}_+^{\varepsilon}(A) \end{aligned}$$

(Definition 1.5.4)

It is appropriate to call these induced functors the *Scharlau transfer functor for Hermitian spaces over involution rings*.

**Definition 1.8.6** ([?, Chapter I, (7.2.4) Proposition]). Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). An *involution trace of  $\eta$*  is a morphism  $\tau : A' \rightarrow A$  of  $A$ -bimodules such that

- $\tau(ayb) = a\tau(y)b$  for all  $a, b \in A$  and  $y \in A'_{(A)}$ , i.e.  $\tau : A'_{(A)} \rightarrow A$  is a bimodule homomorphism.
- 

$$\tau(\bar{a}) = \overline{\tau(a)}$$

for all  $a \in A'$

- The 1-hermitian form  $t_{A'/A} : A' \times A' \rightarrow A$  given by

$$t_{A'/A}(x, y) = \tau(\bar{x}y)$$

is nonsingular (Definition 1.2.3).

**Proposition 1.8.7** ([?, Chapter I, (7.2.4) Proposition]). Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6), and assume that  $A'$  is finitely generated and projective as a right  $A$ -module. Given a right  $A'$ -module  $M'$ , write  $M'_{(A)}$  for the restriction of  $M'$  as a right  $A$ -module (via  $\eta$ ).

Let  $\tau : A' \rightarrow A$  be an involution trace (Definition 1.8.6).

The Scharlau transfer functors (Definition 1.8.5)

$$\begin{aligned} T : \mathfrak{Herm}^{\varepsilon'}(A) &\rightarrow \mathfrak{Herm}^{\varepsilon}(A) \\ T : \mathfrak{Herm}_+^{\varepsilon'}(A) &\rightarrow \mathfrak{Herm}_+^{\varepsilon}(A) \end{aligned}$$

induced by  $\tau$  in turn induce functors

$$\begin{aligned} T : \mathfrak{H}^{\varepsilon'}(A') &\rightarrow \mathfrak{H}^{\varepsilon}(A) \\ T : \mathfrak{H}_+^{\varepsilon'}(A') &\rightarrow \mathfrak{H}_+^{\varepsilon}(A) \end{aligned}$$

(Definition 1.5.14) which may also be called *Scharlau transfer functors*.

**Definition 1.8.8.** (♠ TODO: I don't think I genuinely use this for anything yet) Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Let  $\tau : A' \rightarrow A$  be an involution trace (Definition 1.8.6) of  $\eta$ .

1. We say that  $(A', \tau)$  is a *left perfect trace duality for  $A$  (with respect to  $\eta$ )* if the map

$$\Phi : A' \rightarrow (A')^\vee = \text{Hom}_{\text{Mod}_A}(A', A), \quad b \mapsto (\lambda \mapsto \tau(b \cdot \bar{\lambda}))$$

is an isomorphism of  $(A', A)$ -bimodules. In particular, such a  $\Phi$  is a left Frobenius ring homomorphism (Definition 1.9.1).

2. We say that  $(A', \tau)$  is a *right perfect trace duality for  $A$  (with respect to  $\eta$ )* if the map

$$\Phi : A' \rightarrow (A')^\vee = \text{Hom}_{A \text{ Mod}}(A', A), \quad b \mapsto (\lambda \mapsto \tau(\bar{\lambda} \cdot b))$$

is an isomorphism of  $(A, A')$ -bimodules. In particular, such a  $\Phi$  is a right Frobenius ring homomorphism (Definition 1.9.1).

3.  $(A', \tau)$  is a *perfect trace duality* if it is both left and right perfect.

Note that the maps  $\Phi$  are left/right Frobenius ring homomorphisms (Definition 1.9.1).

**Lemma 1.8.9.** Let  $A$  be a ring with involution (Definition 1.0.5). Let  $\eta : A \rightarrow A'$  be an  $A'$ -algebra with involution (Definition 1.0.8). For any sesquilinear form (Definition 1.1.1)  $b : M \times M \rightarrow A'$  on a (right)  $A'$ -module  $M$ , we have

$$b(ma, n) = b(m, n\bar{a})$$

for all  $m, n \in M$  and  $a \in A$ .

*Proof.* By the sesquilinearity of  $b$  and the fact that  $\eta$  maps  $A$  into the center of  $A'$ , we have

$$b(ma, n) = \bar{a} \cdot b(m, n) = b(m, n) \cdot \bar{a} = b(m, n \cdot \bar{a}).$$

□

**Lemma 1.8.10.** Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Let  $\tau : A' \rightarrow A$  be an involution trace (Definition 1.8.6) of  $\eta$ . Let  $\varepsilon \in Z(A')$  (Definition 1.0.1) such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

1. Suppose that there is some  $u \in A$  such that  $\tau(\varepsilon \cdot \bar{x}) = u \cdot \tau(x)$  for all  $x \in A'$ . We have (♠ TODO: prove basic facts for the purpose of trying to figure out what properties  $\omega = \tau \circ H$  must satisfy in the definition of  $\tau$ - $\varepsilon$ -Hermitian forms assuming this condition)
  - (a)  $\tau(\bar{\varepsilon} \cdot \bar{x}) = \bar{u} \cdot \tau(x)$  for all  $x \in A'$ .
  - (b)  $u^2 = 1$ .
2. Suppose that there is some  $u \in A$  such that  $\tau(\varepsilon \cdot x) = u \cdot \tau(x)$  for all  $x \in A'$ . We have
  - (a)  $\tau(\bar{\varepsilon} \cdot x) = \bar{u} \cdot \tau(x)$  for all  $x \in A'$
  - (b)  $u \cdot \bar{u} = 1$ .

*Proof.* 1. (a) By the fact that  $\tau(\bar{a}) = \overline{\tau(a)}$  for any  $a \in A'$  and the fact that  $\varepsilon$  is in the center of  $A'$ , we have

$$\tau(\bar{\varepsilon} \cdot \bar{x}) = \tau(\bar{\varepsilon} \cdot \bar{x}) = \overline{\tau(\varepsilon \cdot x)} = \overline{u \cdot \tau(x)} = \bar{u} \cdot \overline{\tau(x)} = \bar{u} \cdot \tau(x).$$

(b) For any  $x \in A'$ , we have

$$\tau(x) = \tau(\varepsilon \cdot \bar{\varepsilon} \cdot x) = u \cdot \tau(\varepsilon \bar{x}) = u^2 \cdot \tau(x).$$

Since  $\tau$  is an involution trace, the 1-Hermitian form

$$t_{A'/A} : A' \times A' \rightarrow A, \quad (x, y) \mapsto \tau(\bar{x}y)$$

is nonsingular, i.e. that the adjoint map

$$\hat{t}_{A'/A} : A' \rightarrow (A')^*$$

is an isomorphism of  $A$ -modules. That  $\tau(x) = u^2 \cdot \tau(x)$  holds for every  $x \in A'$  implies that  $(1 - u^2) \cdot t_{A'/A}$  is identically 0 and hence that the composition

$$A' \xrightarrow{t_{A'/A}} (A')^* \xrightarrow{\times(1-u^2)} (A')^*$$

is 0. This can only happen if  $1 = u^2$ .

2. (a) We have

$$\tau(\bar{\varepsilon} \cdot x) = \overline{\tau(\varepsilon \cdot \bar{x})} = \overline{u \cdot \tau(\bar{x})} = \bar{u} \cdot \overline{\tau(\bar{x})} = \bar{u} \cdot \tau(x).$$

(b) For any  $x \in A'$ , we have

$$\tau(x) = \tau(\varepsilon \cdot \bar{\varepsilon} \cdot x) = u \cdot \tau(\bar{\varepsilon} \cdot x) = u \cdot \bar{u} \cdot \tau(x).$$

Similarly as the argument of an above part, the nonsingularity of  $t_{A'/A}$  implies that  $u \cdot \bar{u} = 1$ .

□

**Lemma 1.8.11.** Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Let  $\tau : A' \rightarrow A$  be an involution trace (Definition 1.8.6) of  $\eta$ . Let  $\varepsilon \in Z(A')$  (Definition 1.0.1) such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ . For a  $\varepsilon$ -Hermitian form  $H : M \times M \rightarrow A'$  on a (right)  $A'$ -module  $M$ , write  $\omega = \tau \circ H : M \times M \rightarrow A$  for the image of  $H$  under the Scharlau transfer (Definition 1.8.1) of  $\tau$ ; note that  $\omega$  is sesquilinear (Definition 1.5.1) over  $A$  (Lemma 1.8.2).

1. Suppose that  $u \in A$  satisfies  $\tau(\varepsilon \bar{x}) = u \cdot \tau(x)$  for all  $x \in A'$ ; by Lemma 1.8.10,  $u^2 = 1$ . We have

$$\omega(n, m) = u \cdot \omega(m, n)$$

for all  $n, m \in M$ .

2. Suppose that  $u \in A$  satisfies  $\tau(\varepsilon x) = u \cdot \tau(x)$  for all  $x \in A'$ ; by Lemma 1.8.10,  $u \cdot \bar{u} = 1$ . We have

$$\omega(n, m) = u \cdot \overline{\omega(m, n)}$$

for all  $n, m \in M$ . Assuming that  $u \in Z(A)$ , the form  $\omega$  is a  $u$ -Hermitian form over  $A$ .

*Proof.* 1. For all  $n, m \in M$ ,

$$\omega(n, m) = \tau(H(n, m)) = \tau(\varepsilon \overline{H(m, n)}) = u \cdot \tau(H(m, n)) = u \cdot \omega(m, n).$$

2. For all  $n, m \in M$ , use Lemma 1.8.10 to see that

$$\begin{aligned}\omega(n, m) &= \tau(H(n, m)) = \tau(\varepsilon \cdot \overline{H(m, n)}) = \overline{\tau(\varepsilon \cdot H(m, n))} = \overline{(\bar{u} \cdot \tau(H(m, n)))} = u \cdot \overline{\tau(H(m, n))} \\ &= u \cdot \overline{\omega(m, n)}.\end{aligned}$$

In particular, if  $u \in Z(A)$ , then  $\omega$  is a  $u$ -Hermitian form over  $A$ .

□

**Lemma 1.8.12.** Let  $\eta : A \rightarrow A'$  be a homomorphism of rings with involution (Definition 1.0.6). Let  $b : M \times M \rightarrow A'$  be a sesquilinear form (Definition 1.1.1). Let  $\tau : A' \rightarrow A$  be any function with the trace property (Definition 1.9.7). The composition  $\omega = \tau \circ b$  is (♠ **TODO: balanced**) balanced with respect to  $A'$  in the sense of satisfying

$$\omega(ma, n) = \omega(m, n\bar{a})$$

for any  $m, n \in M$  and  $a \in A'$ .

*Proof.* By the sesquilinearity of  $b$  and the trace property of  $\tau$ , we have

$$\omega(ma, n) = \tau(b(ma, n)) = \tau(\bar{a} \cdot b(m, n)) = \tau(b(m, n) \cdot \bar{a}) = \tau(b(m, n\bar{a})) = \omega(m, n\bar{a}).$$

□

## 1.9. Symmetric algebras with involution rings.

**Definition 1.9.1.** Let  $\varphi : R \rightarrow A$  be a ring homomorphism. We say that  $\varphi$  is a *left (resp. right) Frobenius ring homomorphism* or *left selj-dual* if there exists a left (resp. right)  $A$ -module isomorphism

$$\psi : A \xrightarrow{\cong} A^\vee = \text{Hom}_{R\text{-Mod}}(A, R).$$

(resp.

$$\psi : A \xrightarrow{\cong} A^\vee = \text{Hom}_{\text{Mod}_R}(A, R).$$

) (Definition 1.1.7)

**Lemma 1.9.2.** Let  $\varphi : R \rightarrow A$  be a ring homomorphism.

$\varphi$  is a left (resp. right) Frobenius ring homomorphism (Definition 1.9.1) if and only if there exists a homomorphism  $\lambda : A \rightarrow R$  of left (resp. right)  $R$ -modules such that the map  $B : A \times A \rightarrow R$  defined by  $B(x, y) = \lambda(xy)$

1. is bi-additive,
2. satisfies  $B(xy, z) = B(x, yz)$  for all  $x, y, z \in A$
3. is non-singular, i.e. the mapping

$$A \rightarrow \text{Hom}_{R\text{-Mod}}(A, R), \quad y \mapsto B(\cdot, y)$$

(resp.

$$A \rightarrow \text{Hom}_{\text{Mod}_R}(A, R), \quad x \mapsto B(x, \cdot)$$

) is a left (resp. right)  $A$ -module isomorphism.

*Proof.* Suppose that  $\varphi$  is a left Frobenius ring homomorphism. Given a left  $A$ -module isomorphism  $\psi : A \rightarrow A^\vee = \text{Hom}_{R\text{-Mod}}(A, R)$ , define  $\lambda : A \rightarrow R$  by  $\lambda(a) = \psi(1)(a)$ . It is an element of  $\text{Hom}_{R\text{-Mod}}(A, R)$ , i.e. is a left  $R$ -module homomorphism. Clearly, the induced map  $B$  is bi-additive and satisfies  $B(xy, z) = B(x, yz)$ . Moreover, the map  $y \mapsto B(\cdot, y)$  is exactly the map  $\psi$ , so it is a left  $A$ -module isomorphism.

Conversely, given a left  $R$ -module homomorphism  $\lambda : A \rightarrow R$  satisfying the properties, let  $\psi : A \rightarrow \text{Hom}_{R\text{-Mod}}(A, R)$  be given by  $y \mapsto (x \mapsto B(x, y) = \lambda(x \cdot y))$ . By assumption, this is a left  $A$ -module isomorphism.

The equivalence in the case that  $\varphi$  is a right Frobenius ring homomorphism holds similarly.  $\square$

**Definition 1.9.3** (Frobenius  $R$ -algebra). Let  $R$  be a ring and  $A$  be an  $R$ -algebra.

- $A$  is a *left Frobenius  $R$ -algebra* if the ring homomorphism  $R \rightarrow A$  is a left Frobenius ring homomorphism (Definition 1.9.1).
- $A$  is a *right Frobenius  $R$ -algebra* if the ring homomorphism  $R \rightarrow A$  is a right Frobenius ring homomorphism (Definition 1.9.1).

Some writers will additionally require  $A$  to be a finitely generated and projective module over  $R$ .

For an  $R$ -algebra  $A$  that is finitely generated and projective over  $R$ , left Frobenius  $R$ -algebras are equivalent to right Frobenius  $R$ -algebras. If a "left" isomorphism  $\psi_L$  exists, the corresponding trace  $\varphi = \psi_L(1)$  can always be used to construct a "right" isomorphism  $\psi_R$ .

**Theorem 1.9.4.** Let  $R$  be a ring and let  $\varphi : R \rightarrow A$  be an  $R$ -algebra. Suppose that  $A$  is a left Frobenius  $R$ -algebra (Definition 1.9.3), i.e., there exists an isomorphism of left  $A$ -modules  $\psi : A \rightarrow \text{Hom}_{R\text{-Mod}}(A, R)$ . Let  $\lambda = \psi(1) \in \text{Hom}_{R\text{-Mod}}(A, R)$ . There exists a unique ring automorphism  $\sigma : A \rightarrow A$ , called the *Nakayama automorphism*, such that for all  $x, y \in A$ ,

$$\lambda(xy) = \lambda(y\sigma(x)).$$

Moreover,  $\sigma$  is an  $R$ -algebra automorphism (i.e.,  $\sigma(\varphi(r)) = \varphi(r)$  for all  $r \in R$ ). There exists a similar ring automorphism if  $A$  is a right Frobenius  $R$ -algebra.

*Proof.* (♠ TODO: need to prove uniqueness of  $\sigma$ ) Choose a left  $A$ -module isomorphism  $\psi : A \rightarrow \text{Hom}_{R\text{-Mod}}(A, R)$ . Let  $\lambda = \psi(1) \in \text{Hom}_{R\text{-Mod}}(A, R)$ . For any  $x \in A$ , define a map  $f_x : A \rightarrow R$  by  $f_x(y) = \lambda(xy)$ . Note that  $f_x$  is a left  $R$ -module homomorphism — since  $R$  is central in  $A$ ,

$$f_x(ry) = \lambda(xry) = \lambda(rxy) = r\lambda(xy) = rf_x(y).$$

Define  $\sigma : A \rightarrow A$  by  $\sigma(x) = \psi^{-1}(f_x)$ , i.e.  $\psi(\sigma(x)) = f_x$ . For all  $y \in A$ , since  $\psi$  is an  $A$ -module homomorphism, we have

$$\psi(\sigma(x))(y) = (\sigma(x)\psi(1))(y) = \psi(1)(y \cdot \sigma(x)) = \lambda(y\sigma(x)).$$

On the other hand,

$$\psi(\sigma(x))(y) = f_x(y) = \lambda(xy)$$

by definition, so  $\lambda(xy) = \lambda(y\sigma(x))$ .

We show that  $\sigma$  is an  $R$ -algebra endomorphism. It is additive because

$$\begin{aligned}\psi(\sigma(x_1 + x_2))(y) &= f_{x_1+x_2}(y) = \lambda((x_1 + x_2)(y)) = \lambda(x_1y + x_2y) = \lambda(x_1y) + \lambda(x_2y) \\ &= f_{x_1}(y) + f_{x_2}(y) = \psi(\sigma(x_1))(y) + \psi(\sigma(x_2))(y).\end{aligned}$$

It is multiplicative because, that  $\psi$  is a left  $A$ -module isomorphism and the algebraic property of  $\sigma$  yield

$$\begin{aligned}\psi(\sigma(x_1x_2))(y) &= f_{x_1x_2}(y) = \lambda(x_1x_2y) \\ \psi(\sigma(x_1)\sigma(x_2))(y) &= (\sigma(x_1) \cdot \psi(\sigma(x_2)))(y) = (\sigma(x_1) \cdot f_{x_2})(y) = f_{x_2}(y\sigma(x_1)) = \lambda(x_2y\sigma(x_1)) \\ &= \lambda((x_1y)\sigma(x_1)) = \lambda(x_1(x_2y)) = \lambda(x_1x_2y).\end{aligned}$$

To see that  $\sigma$  is  $R$ -linear, i.e.  $\sigma(r) = r$  for all  $r \in R$ , note that the centrality of  $\varphi$  yields

$$\lambda(y\sigma(r)) = \lambda(ry) = \lambda(yr),$$

for all  $y \in A$ , so

$$\lambda(y(\sigma(r) - r)) = 0.$$

Since this holds for all  $y \in A$  and  $\lambda$  is image of 1 under the isomorphism  $\psi$ , the value  $\sigma(r) - r$  must in fact be 0.

By construction,  $\sigma$  is the composition

$$A \xrightarrow{x \mapsto f_x} \text{Hom}_{R\text{Mod}}(A, R) \xrightarrow{\psi^{-1}} A.$$

Note that  $x \mapsto f_x$  is a bijection by Lemma 1.9.2. Therefore,  $\sigma$  is the composition of bijections and hence is itself a bijection. We have thus shown that  $\sigma$  is an  $R$ -algebra endomorphism.  $\square$

**Theorem 1.9.5.** (♠ TODO: verify; AI generated) Let  $R$  be a ring, and let  $\varphi : R \rightarrow A$  be an  $R$ -algebra that is finitely generated and projective as an  $R$ -module.

Then  $\varphi$  is a left Frobenius ring homomorphism (Definition 1.9.1) if and only if it is a right Frobenius ring homomorphism.

*Proof.* Suppose that  $\varphi$  is a left Frobenius ring homomorphism. By Lemma 1.9.2, there exists a left  $R$ -linear map  $\lambda : A \rightarrow R$  such that the map

$$\Phi_L : A \rightarrow \text{Hom}_{R\text{Mod}}(A, R), \quad y \mapsto (x \mapsto \lambda(xy))$$

is an isomorphism of left  $A$ -modules, where the left  $A$ -action on  $\text{Hom}_{R\text{Mod}}(A, R)$  is given by  $(a \cdot f)(x) = f(xa)$ .

Since  $\varphi(R) \subseteq Z(A)$ , for any  $x \in A$  and  $r \in R$ , we have

$$\lambda(xr) = \lambda(\varphi(r)x) = r\lambda(x) = \lambda(x)r,$$

so  $\lambda$  is also right  $R$ -linear. Define a bi-additive form  $B : A \times A \rightarrow R$  by  $B(x, y) = \lambda(xy)$ . Both variables are  $R$ -linear (left and right), since for all  $r \in R$ :

$$\begin{aligned}B(rx, y) &= \lambda((rx)y) = r\lambda(xy) = rB(x, y), \\ B(x, ry) &= \lambda(x(ry)) = \lambda(\varphi(r)xy) = r\lambda(xy) = rB(x, y).\end{aligned}$$

Define  $\Phi_R : A \rightarrow \text{Hom}_{\text{Mod}_R}(A, R)$  by  $\Phi_R(y)(x) = B(y, x) = \lambda(yx)$ . For  $r \in R$ :

$$\Phi_R(y)(xr) = \lambda(yxr) = \lambda(\varphi(r)yx) = r\lambda(yx) = \Phi_R(y)(x)r,$$

so  $\Phi_R(y)$  is right  $R$ -linear. With the right  $A$ -action on  $\text{Hom}_{\text{Mod}_R}(A, R)$  given by  $(f \cdot a)(x) = f(ax)$ , we have

$$\Phi_R(ya)(x) = \lambda(yax) = \Phi_R(y)(ax) = (\Phi_R(y) \cdot a)(x),$$

so  $\Phi_R$  is right  $A$ -linear.

Since  $A$  is finitely generated projective over  $R$ , the evaluation map  $A \rightarrow A^{**} = \text{Hom}_R \text{Mod}(\text{Hom}_{\text{Mod}_R}(A, R), R)$  is an isomorphism, and non-degeneracy of  $B$  on the left (given by  $\Phi_L$  being an isomorphism) implies non-degeneracy on the right. Hence  $\Phi_R$  is bijective. Thus  $\varphi$  is a right Frobenius ring homomorphism.

The converse direction is symmetric. □

**Definition 1.9.6.** Let  $\varphi : R \rightarrow A$  be a ring homomorphism. We say that  $A$  is *symmetrically self-dual over  $R$  (with respect to  $\varphi$ )* if there exists an isomorphism  $\psi : A \cong A^\vee = \text{Hom}_{\text{Mod}_R}(A, R)$  (Definition 1.1.7) as  $(A, A)$ -bimodules. We may refer to such a  $\psi$  as a *symmetric self-dual map for  $\varphi$* . With respect to a fixed choice of a symmetric self-dual map  $\psi$ , the element  $\psi(1) \in A^\vee$  is called the *symmetric trace*, and satisfies the trace property (Definition 1.9.7) (Lemma 1.9.8).

Clearly, any symmetric self-dual ring homomorphism is a left and a right Frobenius ring homomorphism (Definition 1.9.1).

**Definition 1.9.7.** Let  $f : R \rightarrow S$  be a function from a ring to a set. We say that it satisfies the *trace property* if

$$f(ab) = f(ba)$$

for all  $a, b \in R$ .

**Lemma 1.9.8.** Let  $\varphi : R \rightarrow A$  be a ring homomorphism with respect to which  $A$  is symmetrically self-dual (Definition 1.9.6), i.e. there exists an  $(A, A)$ -bimodule isomorphism  $\psi : A \rightarrow A^\vee = \text{Hom}_R(A, R)$  (Definition 1.1.7). For any  $x \in Z(A)$  (Definition 1.0.1), the functional  $\psi(x)$  is a trace function, i.e. satisfies  $\psi(x)(ab) = \psi(x)(ba)$  for all  $a, b \in A$ .

*Proof.* Note that the  $(A, A)$ -bimodule structure on  $A^\vee$  is defined for  $f \in A^\vee$  and  $u, v, a \in A$  by:

$$\begin{aligned} (u \cdot f)(a) &= f(au) \\ (f \cdot v)(a) &= f(va). \end{aligned}$$

Since  $\psi$  is an  $(A, A)$ -bimodule homomorphism, we have

$$\begin{aligned} \psi(ax) &= a \cdot \psi(x) \\ \psi(xa) &= \psi(x) \cdot a \end{aligned}$$

for all  $a \in A$  and  $x \in Z(A)$ . Therefore, for all  $a, b \in A$  and  $x \in Z(A)$ , we have

$$\psi(x)(ab) = (\psi(x) \cdot a)(b) = (a \cdot \psi(x))(b) = \psi(x)(ba)$$

as desired. □

**Definition 1.9.9** (cf. [?, Chapter I, 7.4] in the case that the base ring is commutative.). Let  $R$  be a not necessarily commutative ring. An  $R$ -algebra  $A$  is called a *symmetric  $R$ -algebra* if the ring homomorphism  $R \rightarrow A$  is symmetrically self-dual over  $R$  (Definition 1.9.6). Many texts additionally require  $A$  to be finitely generated and projective as an  $R$ -module; we do not impose this condition here. Any symmetric  $R$ -algebra is both a left and right Frobenius ring homomorphism (Definition 1.9.1).

**Definition 1.9.10.** Let  $R$  be an involution ring (Definition 1.0.5). A symmetric  $R$ -algebra (Definition 1.9.9)  $A$  is said to be a *symmetric  $R$ -algebra with involution* if

- $A$  has an  $R$ -linear involution (Definition 1.0.8), and (♠ TODO: in Knus, symmetric algebra with involution is formulated in terms of  $\psi(1) \in A^*$ ; the underlying sets of  $A^*$  and  $A^\vee$  should be the same thing, so I think that  $\psi(1)$  being an involution trace should not depend on whether I'm looking at  $A^*$  or  $A^\vee$ . Verify this.)
- the symmetric trace (Definition 1.9.6)  $\psi(1) \in A^\vee$  (Definition 1.1.7) with respect to a chosen symmetric duality  $\psi : A \rightarrow A^\vee = \text{Hom}_R(A, R)$  is an involution trace (Definition 1.8.6).

**Theorem 1.9.11.** (♠ TODO: I don't think symplectic form is the right thing — should be some type of Hermitian form over  $A$  that is "compatible" with  $A'$  or which "admits"  $A'$ )

Let  $A$  be an involution ring (Definition 1.0.5). Let  $A'$  be a symmetric  $A$ -algebra with involution (Definition 1.9.10). Write  $\tau \in A^\vee = \text{Hom}_R(A, R)$  for symmetric trace with respect to the chosen symmetrical duality  $\psi : A \rightarrow A^\vee$ ; it is an involution trace (Definition 1.8.6) by definition. Let  $\varepsilon \in Z(A')$  (Definition 1.0.1) such that  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

Write  $\mathcal{H}_{A'}^\varepsilon$  for the category of modules over  $A'$  with  $\varepsilon$ -Hermitian forms (Definition 1.2.2).

1. Suppose that there exists  $u \in A$  such that  $\tau(\varepsilon \cdot \bar{x}) = u \cdot \tau(x)$  for all  $x \in A'$ .  
Write  $\mathcal{S}_A^{\text{comp}}(\eta)$  for the category of modules over  $A'$  with sesquilinear forms  $\omega : M \times M \rightarrow A$  over  $A$ 
  - such that

$$\omega(m, n) = u \cdot \omega(n, m)$$

for all  $m, n \in M$  and

- that are compatible with  $A'$  in the sense that

$$\omega(m \cdot a', n) = \omega(m, n \cdot \bar{a}')$$

for all  $a' \in A'$  and  $m, n \in M$ .

There is an equivalence

$$\mathcal{H}_{A'}^\varepsilon \xrightarrow{\sim} \mathcal{S}_A^{\text{comp}}(\eta), \quad H \mapsto \tau \circ H.$$

(♠ TODO: comment on nondegeneracy, finitely generated projectiveness) This equivalence restricts to the subcategory of  $\mathcal{H}_{A'}^\varepsilon(\tau)$  of nondegenerate  $\varepsilon - \tau$ -Hermitian forms.

2. Suppose that there exists  $u \in A$  such that  $\tau(\varepsilon \cdot x) = u \cdot \tau(x)$  for all  $x \in A'$ . Write  $\mathcal{S}_A^{\text{comp}}(\eta)$  for the category of modules over  $A'$  with sesquilinear forms (Definition 1.1.1)  $\omega : M \times M \rightarrow A$  over  $A$ 
  - such that

$$\omega(m, n) = u \cdot \overline{\omega(n, m)}$$

- for all  $m, n \in M$  (note that  $u \cdot \bar{u} = 1$  by Lemma 1.8.10, so if  $u \in Z(A)$ , then  $\omega$  is  $u$ -Hermitian (Definition 1.2.1) over  $A$ ) and
- that are compatible with  $A'$  in the sense that

$$\omega(m \cdot a', n) = \omega(m, n \cdot \bar{a}')$$

for all  $a' \in A'$  and  $m, n \in M$ .

There is an equivalence

$$\mathcal{H}_{A'}^\varepsilon \xrightarrow{\sim} \mathcal{S}_A^{\text{comp}}(\eta), \quad \tau \circ H.$$

This equivalence restricts to the subcategory of  $\mathcal{H}_{A'}^\varepsilon(\tau)$  of nondegenerate  $\varepsilon$ - $\tau$ -Hermitian forms.

(♠ TODO: factor out the notion of "compatible" as its own definition)

(♠ TODO: prove)

*Proof.* Given  $(M, H) \in \mathcal{H}_{A'}^\varepsilon(\tau)$ , let  $\omega_H$  be the form  $\tau \circ H : M \times M \rightarrow A$ . By Lemma 1.9.8, the symmetric trace  $\tau = \psi(1)$  has the trace property (Definition 1.9.7). Therefore,  $\omega_H$  satisfies

$$\omega_H(ma, n) = \omega_H(m, n\bar{a})$$

for all  $a \in A'$  and  $m, n \in M$  by Lemma 1.8.9. Moreover,  $\omega_H(n, m) = u \cdot \omega_H(m, n)$  for all  $n, m \in M$  by Lemma 1.8.11. Therefore, we indeed have a functor in the forward direction.

Given  $(M, \omega) \in \mathcal{S}_A^{\text{comp}}(\eta)$ , and for all  $m, n \in M$ , define  $\phi(m, n) \in \text{Hom}_A(A', A) = (A')^\vee$  by  $\phi(m, n)(a) = \omega(ma, n)$  (Definition 1.1.7). Writing

$$\widehat{t}_{A'/A} : A' \rightarrow (A')^* \cong \overline{(A')^\vee}$$

(Definition 1.1.9) (Definition 1.1.11) for the (left) adjoint map (Definition 1.1.14) of the 1-Hermitian form  $t_{A'/A} : A' \times A' \rightarrow A$  given by the involution trace  $\tau$ , define

$$H_\omega(m, n) = \widehat{t}_{A'/A}^{-1}(\phi(m, n)).$$

We show that it is  $\varepsilon$ -Hermitian (Definition 1.2.1).

For all  $m_1, m_2, n \in M$  and  $a \in M$ , we have

$$\begin{aligned} \widehat{t}_{A'/A}(H_\omega(m_1 + m_2, n))(a) &= \phi(m_1 + m_2, n)(a) = \omega((m_1 + m_2)a, n) = \omega(m_1a, n) + \omega(m_2a, n) \\ &= \widehat{t}_{A'/A}(H_\omega(m_1, n))(a) + \widehat{t}_{A'/A}(H_\omega(m_2, n))(a), \end{aligned}$$

so

$$H_\omega(m_1 + m_2, n) = H_\omega(m_1, n) + H_\omega(m_2, n),$$

i.e.  $H$  is additive in the first argument. Similarly,  $H$  is additive in the second argument, so it is bi-additive. For all  $m, n \in M$  and  $a_1, a_2, b \in A'$ , we have

$$\begin{aligned} \widehat{t}_{A'/A}(H_\omega(ma_1, na_2))(b) &= \phi(ma_1, na_2)(b) = \omega(ma_1b, na_2) = \omega(m(a_1b\bar{a}_2), n) \\ &= \phi(m, n)(a_1b\bar{a}_2) \\ &= (\bar{a}_1\phi(m, n)a_2)(b) \\ &= \widehat{t}_{A'/A}(\bar{a}_1H_\omega(m, n)a_2)(b), \end{aligned}$$

so  $H_\omega$  is sesquilinear. For all  $m, n \in M$ , we also have

$$\begin{aligned}\widehat{t}_{A'/A}(H_\omega(n, m))(b) &= \omega(nb, m) = u \cdot \omega(m, nb) = u \cdot \omega(m \cdot \bar{b}, n) = u \cdot \phi(m, n)(\bar{b}) \\ &= u \cdot \tau(H_\omega(m, n) \cdot \bar{b}) = u \cdot \tau(H_\omega(m, n) \cdot b)\end{aligned}$$

note that we used above that  $\widehat{t}_{A'/A}(H_\omega(m, n))(b) = \phi(m, n)(b) = \tau(H_\omega(m, n) \cdot \bar{b})$ . using the assumption  $\tau(\varepsilon \cdot \bar{x}) = u \cdot \tau(x)$ , we obtain

$$u \cdot \tau(H_\omega(m, n) \cdot b) = \tau(\varepsilon \cdot \overline{H_\omega(n, m) \cdot b}) = \tau(\varepsilon \cdot \bar{b} \cdot \overline{H_\omega(m, n)}).$$

By the trace property  $\tau(xy) = \tau(yx)$ , this equals

$$\tau(\overline{H_\omega(m, n)} \cdot \varepsilon \cdot \bar{b}) = \widehat{t}_{A'/A}(\varepsilon \cdot \overline{H_\omega(m, n)})(b).$$

Therefore,  $H_\omega$  is  $\varepsilon$ -Hermitian. We thus have a functor in the backward direction.

To see that these functors are natural inverses, note that for  $(M, H) \in \mathcal{H}_{A'}^\varepsilon$ , the adjoint definition  $\phi(m, n)(a) = \omega_H(m, n) = \tau(H(ma, n))$  implies that  $H_{\omega_H}(m, n) = H(m, n)$  by the uniqueness of the adjoint map  $\widehat{t}_{A'/A}$ . Conversely, for  $(M, \omega) \in \mathcal{S}_A^{\text{comp}}$ , evaluation of the adjoint at the identity  $a = 1$  yields  $\omega(m, n) = \tau(H_\omega(m, n))$ . (♠ TODO: prove that nondegenerate things are taken to nondegenerate things)

□

**Definition 1.9.12.** Let  $G$  be a group and  $R$  a not necessarily commutative ring.

The *group ring*  $R[G]$ , also denoted  $RG$ , is the set of all formal linear combinations of the form

$$\sum_{g \in G} r_g g$$

where  $r_g \in R$  and  $r_g = 0$  for all but finitely many  $g \in G$ . This set is endowed with a ring structure via:

- Addition:

$$\left( \sum_{g \in G} r_g g \right) + \left( \sum_{g \in G} s_g g \right) = \sum_{g \in G} (r_g + s_g) g$$

- Multiplication: Defined by the extension of the group law and the requirement that  $R$  and  $G$  commute ( $rg = gr$ ):

$$\left( \sum_{g \in G} r_g g \right) \left( \sum_{h \in G} s_h h \right) = \sum_{k \in G} \left( \sum_{gh=k} r_g s_h \right) k$$

The identity element of  $R[G]$  is  $1_R 1_G$ .

$R[G]$  may also be called the *group algebra of  $G$  over  $R$*  as the natural ring homomorphism  $R \rightarrow R[G]$  sending  $r \mapsto r 1_G$  is an algebra

**Definition 1.9.13** (see [?, 7.3.7]). Let  $(R, \sigma_R)$  be a ring with involution (Definition 1.0.5). Let  $G$  be a group.

The *canonical involution on the group ring  $R[G]$*  (Definition 1.9.12) is the  $R$ -linear involution (Definition 1.0.8) given by

$$\overline{\sum r_g g} = \sum \sigma_R(r_g) g^{-1}.$$

If  $R$  is not specified with an involution structure, then usually, the canonical involution on  $R[G]$  refers to the above in the case that  $R$  is equipped with the trivial involution (Definition 1.0.7).

**Definition 1.9.14** (see [?, 7.3.7]). Let  $(R, \sigma_R)$  be a ring with involution (Definition 1.0.5). Let  $G$  be a group.

The *augmentation trace* or *standard trace*  $\tau : R[G] \rightarrow R$  is defined as

$$\tau \left( \sum_{g \in G} a_g g \right) = a_1$$

where 1 is the identity element of  $G$ .

Assuming that  $R$  is commutative and that  $G$  is finite, this is an involution trace (Definition 1.8.6) with respect to the canonical involution (Definition 1.9.13) of  $R[G]$  and is also a symmetric trace (Definition 1.9.6), and hence  $R[G]$  is a symmetric algebra with involution (Definition 1.9.10) over  $R$ .

**Proposition 1.9.15.** Let  $R$  be a commutative ring. Any finite étale  $R$ -algebra  $A$  is a symmetric  $R$ -algebra (Definition 1.9.9) that is finitely generated and projective as an  $R$ -module. In particular, the relative trace map  $\text{Tr}_{A/R} : A \rightarrow R$  is a symmetric trace.

*Proof.* Since  $A$  is a finite  $R$ -algebra,  $A$  is a finitely generated  $R$ -module by definition. Since  $A$  is étale over  $R$ , it is flat over  $R$ . Since  $A$  is also finitely generated over  $R$ , it is projective over  $R$ . The étaleness of  $A$  also implies that it is separable and hence the bilinear form  $A \times A \rightarrow R, (x, y) \mapsto \text{Tr}_{A/T}(xy)$  is nonsingular. The map

$$A \rightarrow A^\vee = \text{Hom}_R(A, R), \quad x \mapsto (y \mapsto \text{Tr}_{A/T}(xy))$$

is thus a symmetric self-dual map (Definition 1.9.6), i.e. is an  $A$ - $A$ -bimodule isomorphism (♠ TODO: why), so  $A$  is a symmetric  $R$ -algebra as desired. (♠ TODO: relative trace map) (♠ TODO: nonsingular)  $\square$

**Proposition 1.9.16.** Let  $R$  be a commutative ring. Let  $G$  be a finite group. The group algebra (Definition 1.9.12)  $R[G]$  is a symmetric  $R$ -algebra (Definition 1.9.9); the augmentation trace (Definition 1.9.14)  $\tau : R[G] \rightarrow R$  is a symmetric trace.

*Proof.* The map  $R[G] \times R[G] \rightarrow R, (x, y) \mapsto \tau(xy)$  is symmetric (♠ TODO: ), and non-degenerate (♠ TODO: make a lemma for this nonsingularity). Thus,  $R[G]$  is a symmetric  $R$ -algebra with symmetric trace  $\tau$ .  $\square$

**Proposition 1.9.17.** Let  $R$  be a commutative ring (regarded as an involution ring (Definition 1.0.5) with the trivial involution (Definition 1.0.7)). Let  $G$  be a finite group. The group algebra (Definition 1.9.12)  $R[G]$  is a symmetric involution algebra (Definition 1.9.10) over  $R$  with the canonical involution (Definition 1.9.13); the augmentation trace (Definition 1.9.14)  $\tau : R[G] \rightarrow R$  is an involution trace (Definition 1.8.6).

*Proof.* By Proposition 1.9.16,  $R[G]$  is a symmetric  $R$ -algebra with  $\tau$  as a symmetric trace (Definition 1.9.6). It remains to show that  $\tau$  is an involution trace. It is clearly an  $R$ -bimodule homomorphism, we have

$$\tau \left( \overline{\sum_{g \in G} a_g g} \right) = \tau \left( \sum_{g \in G} a_g g^{-1} \right) = a_1 = \tau \left( \sum_{g \in G} a_g g \right),$$

(recall that  $R$  is equipped with the trivial involution) and that  $R[G] \times R[G] \rightarrow R$ ,  $(x, y) \mapsto \tau(xy)$  is nonsingular (♠ TODO: ). Thus,  $\tau$  is indeed an involution trace.  $\square$

**Example 1.9.18.** The following are examples of symmetric algebras (Definition 1.9.9):

1. Finite field extensions (♠ TODO: )
2. Finite étale  $R$ -algebras (♠ TODO: )
3. Azumaya  $R$ -algebras (♠ TODO: )
4. Group rings (Definition 1.9.12)  $R[G]$  equipped with the canonical involution (Definition 1.9.13) where  $R$  is commutative ring and a  $G$  is a finite group are symmetric algebras with involution (Definition 1.9.10).

**Definition 1.9.19** (cf. [?, Chapter I. 7.4] in the case that the base ring is commutative). Let  $R$  be a not necessarily commutative ring, and  $A$  a symmetric  $R$ -algebra (Definition 1.9.9) with an  $R$ -linear involution (Definition 1.0.8)  $\sigma : A \rightarrow A$  denoted by  $\bar{a}$ . Let  $N$  be a finitely generated projective right  $A$ -module. We view  $N$  also as an  $(R, A)$ -bimodule via the  $R$ -algebra structure of  $A$ .

An  $R$ -valued form  $k : N \otimes_{\mathbb{Z}} N \rightarrow R$  (which is tantamount to a bilinear form  $N \times N \rightarrow R$  (??)) is said to **admit  $A$**  if it is an  $(R, R)$ -bimodule map that factors through the tensor product over  $A$ . Specifically, it must satisfy:

1. **Balanced over  $A$ :** For all  $x, y \in N$  and  $a \in A$ ,

$$k(xa, y) = k(x, ay)$$

where the right action of  $A$  on the left side is the module action, and the left action of  $A$  on the right side is defined via the involution:  $ay := y\bar{a}$ .

2. **Compatibility with  $R$ -structure:** For all  $r_1, r_2 \in R$ ,

$$k(r_1x, yr_2) = r_1k(x, y)r_2$$

This is equivalent to  $k$  inducing a well-defined  $(R, R)$ -bimodule homomorphism:

$$\tilde{k} : N \otimes_A \overline{N} \longrightarrow R$$

where  $\overline{N}$  is the opposite (Definition 1.1.9) of  $N$  as a right module over the involution ring  $A$ .

**Definition 1.9.20** (cf. [?, Chapter I, 7.4] for a discussion in the case where the base ring is commutative). Let  $R$  be a not necessarily commutative ring, let  $\varepsilon \in R^\times$  satisfy  $\varepsilon^2 = 1$ , and  $A$  a symmetric  $R$ -algebra (Definition 1.9.9) with an  $R$ -linear involution (Definition 1.0.8).

The *category of  $\varepsilon$ -Hermitian spaces over  $R$  which admit  $A$*  is the full subcategory  $\mathfrak{H}^\varepsilon(A)$  of the category of modules with  $\varepsilon$ -Hermitian forms over  $R$  (Definition 1.2.2) whose objects are  $\varepsilon$ -Hermitian spaces (Definition 1.5.13) over  $R$  whose underlying Hermitian forms admit  $A$  (Definition 1.9.19). In particular, such a category is a groupoid.

The *category of even  $\varepsilon$ -Hermitian spaces over  $R$  which admit  $A$*  is the full subcategory  $\mathfrak{H}_+^\varepsilon(A)$  whose objects are those of  $\mathfrak{H}^\varepsilon(A)$  whose underlying Hermitian forms are even (Definition 1.4.1).

**Theorem 1.9.21** ([?, Chapter I, 7.4.1 Theorem]). Let  $R$  be a commutative ring. Let  $A$  be a symmetric  $R$ -algebra with involution (Definition 1.9.10) that is finitely generated and projective as an  $R$ -module. Let  $\varepsilon \in A$  be an element satisfying  $\varepsilon \cdot \bar{\varepsilon} = 1$ .

The Scharlau transfer functors (Definition 1.8.5)

$$\begin{aligned} T : \mathfrak{Herm}^\varepsilon(A) &\rightarrow \mathfrak{Herm}^\varepsilon(R) \\ T : \mathfrak{Herm}_+^\varepsilon(A) &\rightarrow \mathfrak{Herm}_+^\varepsilon(R) \end{aligned}$$

give rise to isomorphisms of categories

$$\begin{aligned} \mathfrak{H}^\varepsilon(A) &\xrightarrow{\sim} \mathfrak{H}^\varepsilon(R; A) \\ \mathfrak{H}_+^\varepsilon(A) &\xrightarrow{\sim} \mathfrak{H}_+^\varepsilon(R; A) \end{aligned}$$

(Definition 1.5.14) (Definition 1.9.20) which preserve orthogonal sums and hyperbolic spaces.  
(♠ TODO: orthogonal sums, hyperbolic spaces)

## 1.10. Involution, sesquilinearity, hermitian form over sheaves of rings.

**Definition 1.10.1.** Let  $(\mathcal{S}, \tau)$  be a site. Let  $\mathcal{O}$  be a sheaf of rings on  $\mathcal{S}$ . An *involution on  $\mathcal{O}$*  is a morphism of sheaves of rings  $\sigma : \mathcal{O} \rightarrow \mathcal{O}^{\text{op}}$  such that  $\sigma \circ \sigma = \text{id}_{\mathcal{O}}$ , i.e. an involution (Definition 1.0.3) in the category of sheaves of rings on  $\mathcal{S}$ .

**Definition 1.10.2.** Let  $(\mathcal{S}, \tau)$  be a site. Let  $(\mathcal{O}, \sigma)$  be a sheaf of rings on  $\mathcal{S}$  with involution (Definition 1.10.1)  $\sigma : \mathcal{O} \rightarrow \mathcal{O}$ . Let  $\mathcal{E}$  be a sheaf of left  $\mathcal{O}$ -modules on  $\mathcal{S}$ .

A *sesquilinear form on  $\mathcal{E}$*  is a morphism of sheaves of abelian groups

$$\phi : \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{O}$$

such that for every object  $U \in \mathcal{S}$ , the induced map on sections

$$\phi_U : \mathcal{E}(U) \times \mathcal{E}(U) \rightarrow \mathcal{O}(U)$$

is a  $\sigma_U$ -sesquilinear form (Definition 1.1.1) on the  $\mathcal{O}(U)$ -module  $\mathcal{E}(U)$ , where  $\sigma_U : \mathcal{O}(U) \rightarrow \mathcal{O}(U)$  is the involution (Definition 1.0.5) on global sections.

**Definition 1.10.3.** Let  $(\mathcal{O}, \sigma)$  be a sheaf of rings with involution (Definition 1.10.1) on a site  $\mathcal{S}$ . Let  $\mathcal{E}$  be a sheaf of left  $\mathcal{O}$ -modules.

A *hermitian form on  $\mathcal{E}$*  is a morphism of sheaves of sets

$$\phi : \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{O}$$

such that for every object  $U$  in  $\mathcal{S}$ , the map on sections

$$\phi_U : \mathcal{E}(U) \times \mathcal{E}(U) \rightarrow \mathcal{O}(U)$$

is a hermitian form (Definition 1.2.1) on the  $\mathcal{O}(U)$ -module  $\mathcal{E}(U)$  with respect to the involution (Definition 1.0.5)  $\sigma_U : \mathcal{O}(U) \rightarrow \mathcal{O}(U)$ .